

**IT3021**

**Data Warehousing & Business Intelligence**

**Assignment 1**

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**IT23541184**

# 1. Data set selection

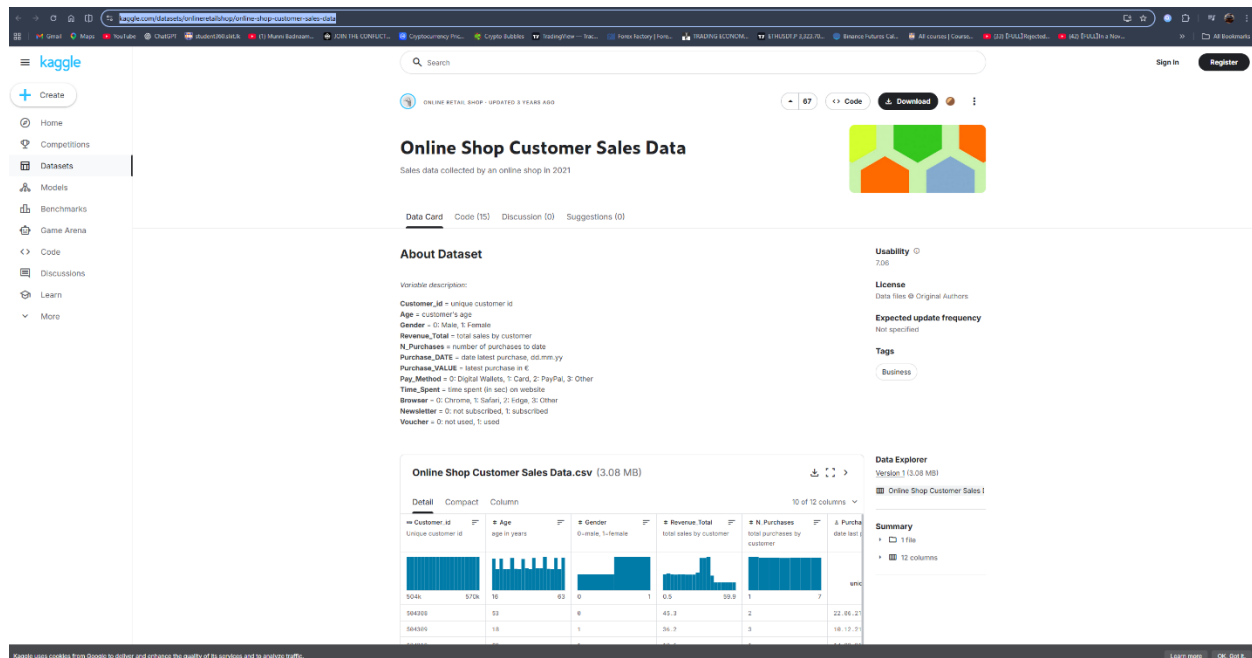
I have selected **Online Shop Customer Sales Data** dataset for the assignment.

## 1.1.Data Set: **Online Shop Customer Sales Data**

<https://www.kaggle.com/datasets/onlinetailshop/online-shop-customer-sales-data>

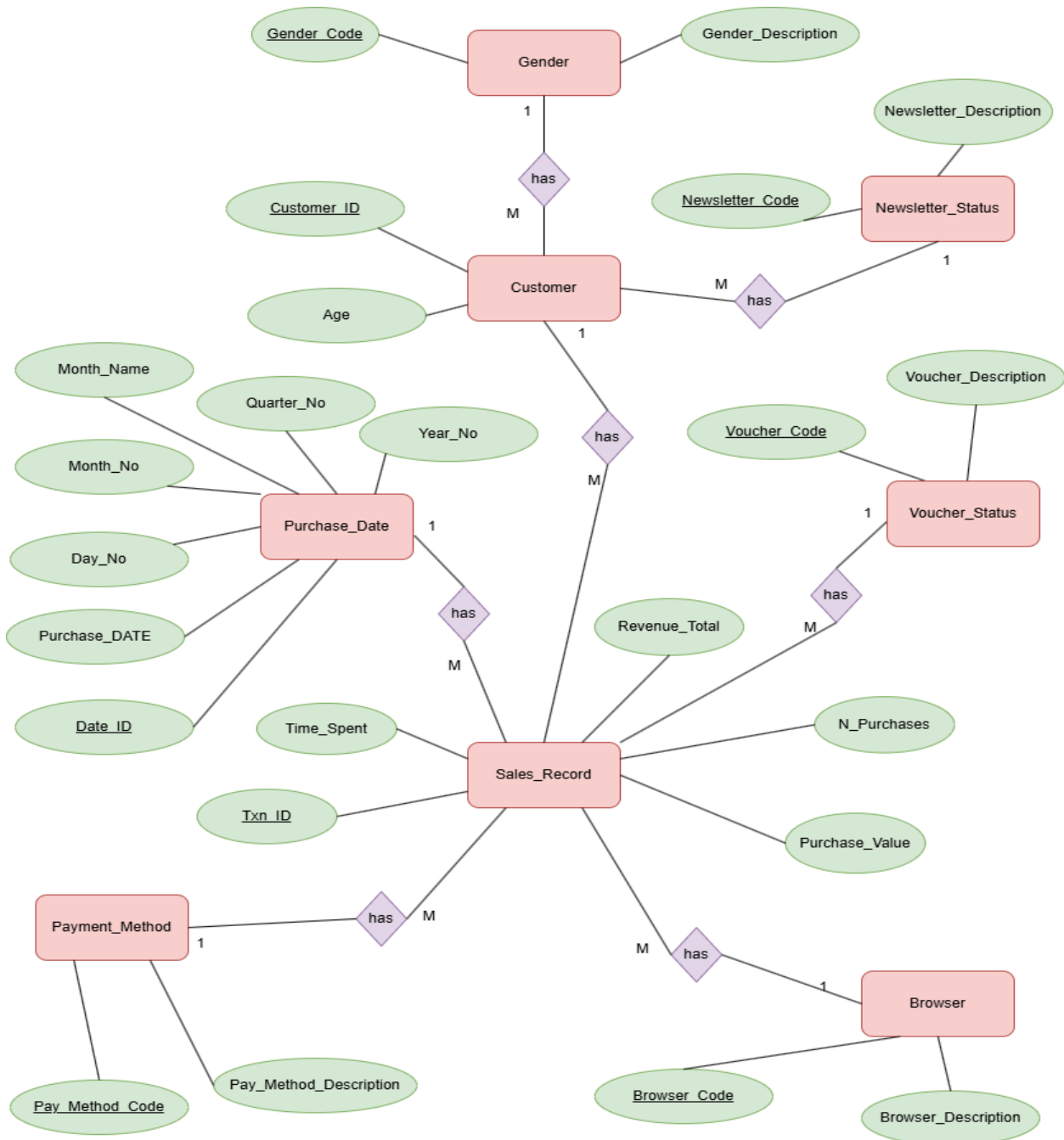
The dataset selected for this assignment is the **Online Shop Customer Sales Data** dataset. It contains detailed operational records related to online retail sales, including attributes such as **Customer\_id**, **Age**, **Gender**, **Revenue\_Total**, **N\_Purchases**, **Purchase\_DATE**, **Purchase\_VALUE**, **Pay\_Method**, **Time\_Spent**, **Browser**, **Newsletter**, and **Voucher**. Each row captures a single sales-related event at record level. As a result, the dataset is suitable to be treated as an **OLTP dataset**, since it stores detailed source-level operational data rather than a pre-built dimensional model.

The main source file used for this assignment is **Online Shop Customer Sales Data.csv**, which provides the core data required for warehouse design and ETL development. The dataset covers approximately one year of data and contains sufficient detail to support the requirements of the assignment, including multiple source preparation, dimensional modelling, ETL design, hierarchies, dimensions, aggregates, and reporting. The grain of the dataset is defined as **one record per online shop sales record**. To make the dataset more suitable for data warehousing implementation, several modifications were planned during source preparation, such as separating the flat source into multiple source types and introducing supporting structures needed for later ETL and dimensional modelling.



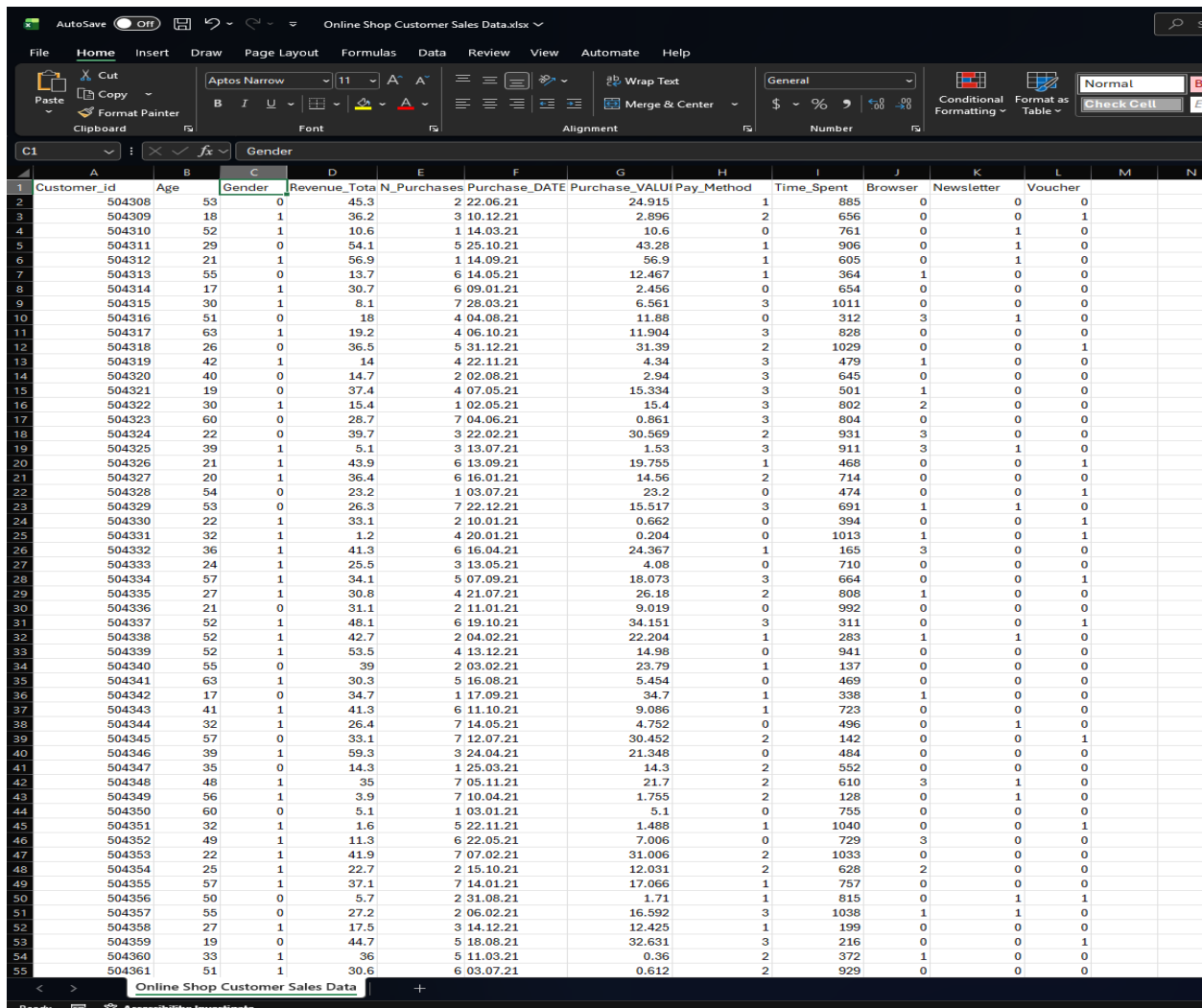
## ER Diagram

The ER diagram was developed by normalizing the flat source CSV into related entities. Since the original dataset does not contain a transaction identifier, a synthetic key Txn\_ID was introduced for the Sales\_Record entity. Customer-related attributes were separated from transaction-related attributes, while coded fields such as gender, payment method, browser, newsletter, and voucher were modeled as lookup/master entities.



## 2: Preparation of data sources

The selected dataset was originally available as a single CSV source containing both customer and sales information. To prepare the data for this assignment, the source was divided into separate and meaningful data sources. The final prepared sources are Customer Details, Sales Record Details, and Lookup Details. Customer details include Customer\_id, Age, Gender, and Newsletter, while sales record details include Customer\_id, Purchase\_DATE, Revenue\_Total, N\_Purchases, Purchase\_VALUE, Pay\_Method, Time\_Spent, Browser, and Voucher. Lookup details were created to provide descriptive values for coded fields. These prepared sources were then stored in different formats such as CSV, Excel, and SQL Server database tables to satisfy the requirement of having multiple source types.



|    | A           | B   | C      | D             | E           | F             | G              | H          | I          | J       | K          | L       | M | N |
|----|-------------|-----|--------|---------------|-------------|---------------|----------------|------------|------------|---------|------------|---------|---|---|
| 1  | Customer_id | Age | Gender | Revenue_Total | N_Purchases | Purchase_DATE | Purchase_VALUE | Pay_Method | Time_Spent | Browser | Newsletter | Voucher |   |   |
| 2  | 504308      | 53  | 0      | 45.3          | 2           | 22.06.21      | 24.915         | 1          | 885        | 0       | 0          | 0       |   |   |
| 3  | 504309      | 18  | 1      | 36.2          | 3           | 10.12.21      | 2.896          | 2          | 656        | 0       | 0          | 1       |   |   |
| 4  | 504310      | 52  | 1      | 10.6          | 1           | 14.03.21      | 10.6           | 0          | 761        | 0       | 1          | 0       |   |   |
| 5  | 504311      | 29  | 0      | 54.1          | 5           | 25.10.21      | 43.28          | 1          | 906        | 0       | 1          | 0       |   |   |
| 6  | 504312      | 21  | 1      | 56.9          | 1           | 14.09.21      | 56.9           | 1          | 605        | 0       | 1          | 0       |   |   |
| 7  | 504313      | 55  | 0      | 13.7          | 6           | 14.05.21      | 12.467         | 1          | 364        | 1       | 0          | 0       |   |   |
| 8  | 504314      | 17  | 1      | 30.7          | 6           | 09.01.21      | 2.456          | 0          | 654        | 0       | 0          | 0       |   |   |
| 9  | 504315      | 30  | 1      | 8.1           | 7           | 28.03.21      | 6.561          | 3          | 1011       | 0       | 0          | 0       |   |   |
| 10 | 504316      | 51  | 0      | 18            | 4           | 04.08.21      | 11.88          | 0          | 312        | 3       | 1          | 0       |   |   |
| 11 | 504317      | 63  | 1      | 19.2          | 4           | 06.10.21      | 11.904         | 3          | 828        | 0       | 0          | 0       |   |   |
| 12 | 504318      | 26  | 0      | 36.5          | 5           | 31.12.21      | 31.39          | 2          | 1029       | 0       | 0          | 1       |   |   |
| 13 | 504319      | 42  | 1      | 14            | 4           | 22.11.21      | 4.34           | 3          | 479        | 1       | 0          | 0       |   |   |
| 14 | 504320      | 40  | 0      | 14.7          | 2           | 02.08.21      | 2.94           | 3          | 645        | 0       | 0          | 0       |   |   |
| 15 | 504321      | 19  | 0      | 37.4          | 4           | 07.05.21      | 15.334         | 3          | 501        | 1       | 0          | 0       |   |   |
| 16 | 504322      | 30  | 1      | 15.4          | 1           | 02.05.21      | 15.4           | 3          | 802        | 2       | 0          | 0       |   |   |
| 17 | 504323      | 60  | 0      | 28.7          | 7           | 04.06.21      | 0.861          | 3          | 804        | 0       | 0          | 0       |   |   |
| 18 | 504324      | 22  | 0      | 39.7          | 3           | 22.02.21      | 30.569         | 2          | 931        | 3       | 0          | 0       |   |   |
| 19 | 504325      | 39  | 1      | 5.1           | 3           | 13.07.21      | 1.53           | 3          | 911        | 3       | 1          | 0       |   |   |
| 20 | 504326      | 21  | 1      | 43.9          | 6           | 13.09.21      | 19.755         | 1          | 468        | 0       | 0          | 1       |   |   |
| 21 | 504327      | 20  | 1      | 36.4          | 6           | 16.01.21      | 14.56          | 2          | 714        | 0       | 0          | 0       |   |   |
| 22 | 504328      | 54  | 0      | 23.2          | 1           | 03.07.21      | 23.2           | 0          | 474        | 0       | 0          | 1       |   |   |
| 23 | 504329      | 53  | 0      | 26.3          | 7           | 22.12.21      | 15.517         | 3          | 691        | 1       | 1          | 0       |   |   |
| 24 | 504330      | 22  | 1      | 33.1          | 2           | 10.01.21      | 0.662          | 0          | 394        | 0       | 0          | 1       |   |   |
| 25 | 504331      | 32  | 1      | 1.2           | 4           | 20.01.21      | 0.204          | 0          | 1013       | 1       | 0          | 1       |   |   |
| 26 | 504332      | 36  | 1      | 41.3          | 6           | 16.04.21      | 24.367         | 1          | 165        | 3       | 0          | 0       |   |   |
| 27 | 504333      | 24  | 1      | 25.5          | 3           | 13.05.21      | 4.08           | 0          | 710        | 0       | 0          | 0       |   |   |
| 28 | 504334      | 57  | 1      | 34.1          | 5           | 07.09.21      | 18.073         | 3          | 664        | 0       | 0          | 1       |   |   |
| 29 | 504335      | 27  | 1      | 30.8          | 4           | 21.07.21      | 26.18          | 2          | 808        | 1       | 0          | 0       |   |   |
| 30 | 504336      | 21  | 0      | 31.1          | 2           | 11.01.21      | 9.019          | 0          | 992        | 0       | 0          | 0       |   |   |
| 31 | 504337      | 52  | 1      | 48.1          | 6           | 19.10.21      | 34.151         | 3          | 311        | 0       | 0          | 1       |   |   |
| 32 | 504338      | 52  | 1      | 42.7          | 2           | 04.02.21      | 22.204         | 1          | 283        | 1       | 1          | 0       |   |   |
| 33 | 504339      | 52  | 1      | 53.5          | 4           | 13.12.21      | 14.98          | 0          | 941        | 0       | 0          | 0       |   |   |
| 34 | 504340      | 55  | 0      | 39            | 2           | 03.02.21      | 23.79          | 1          | 137        | 0       | 0          | 0       |   |   |
| 35 | 504341      | 63  | 1      | 30.3          | 5           | 16.08.21      | 5.454          | 0          | 469        | 0       | 0          | 0       |   |   |
| 36 | 504342      | 17  | 0      | 34.7          | 1           | 17.09.21      | 34.7           | 1          | 338        | 1       | 0          | 0       |   |   |
| 37 | 504343      | 41  | 1      | 41.3          | 6           | 11.10.21      | 9.086          | 1          | 723        | 0       | 0          | 0       |   |   |
| 38 | 504344      | 32  | 1      | 26.4          | 7           | 14.05.21      | 4.752          | 0          | 496        | 0       | 1          | 0       |   |   |
| 39 | 504345      | 57  | 0      | 33.1          | 7           | 12.07.21      | 30.452         | 2          | 142        | 0       | 0          | 1       |   |   |
| 40 | 504346      | 39  | 1      | 59.3          | 3           | 24.04.21      | 21.348         | 0          | 484        | 0       | 0          | 0       |   |   |
| 41 | 504347      | 35  | 0      | 14.3          | 1           | 25.03.21      | 14.3           | 2          | 552        | 0       | 0          | 0       |   |   |
| 42 | 504348      | 48  | 1      | 35            | 7           | 05.11.21      | 21.7           | 2          | 610        | 3       | 1          | 0       |   |   |
| 43 | 504349      | 56  | 1      | 3.9           | 7           | 10.04.21      | 1.755          | 2          | 128        | 0       | 1          | 0       |   |   |
| 44 | 504350      | 60  | 0      | 5.1           | 1           | 03.01.21      | 5.1            | 0          | 755        | 0       | 0          | 0       |   |   |
| 45 | 504351      | 32  | 1      | 1.6           | 5           | 22.11.21      | 1.488          | 1          | 1040       | 0       | 0          | 1       |   |   |
| 46 | 504352      | 49  | 1      | 11.3          | 6           | 22.05.21      | 7.006          | 0          | 729        | 3       | 0          | 0       |   |   |
| 47 | 504353      | 22  | 1      | 41.9          | 7           | 07.02.21      | 31.006         | 2          | 1033       | 0       | 0          | 0       |   |   |
| 48 | 504354      | 25  | 1      | 22.7          | 2           | 15.10.21      | 12.031         | 2          | 628        | 2       | 0          | 0       |   |   |
| 49 | 504355      | 57  | 1      | 37.1          | 7           | 14.01.21      | 17.066         | 1          | 757        | 0       | 0          | 0       |   |   |
| 50 | 504356      | 50  | 0      | 5.7           | 2           | 31.08.21      | 1.71           | 1          | 815        | 0       | 1          | 1       |   |   |
| 51 | 504357      | 55  | 0      | 27.2          | 2           | 06.02.21      | 16.592         | 3          | 1038       | 1       | 1          | 0       |   |   |
| 52 | 504358      | 27  | 1      | 17.5          | 3           | 14.12.21      | 12.425         | 0          | 199        | 0       | 0          | 0       |   |   |
| 53 | 504359      | 19  | 0      | 44.7          | 5           | 18.08.21      | 32.631         | 3          | 216        | 0       | 0          | 1       |   |   |
| 54 | 504360      | 33  | 1      | 36            | 5           | 11.03.21      | 0.36           | 2          | 372        | 1       | 0          | 0       |   |   |
| 55 | 504361      | 51  | 1      | 30.6          | 6           | 03.07.21      | 0.612          | 2          | 929        | 0       | 0          | 0       |   |   |

Online Shop Customer Sales Data.csv

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Clipboard

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G8

|    | A           | B   | C      | D          | E | F | G |
|----|-------------|-----|--------|------------|---|---|---|
| 1  | Customer_id | Age | Gender | Newsletter |   |   |   |
| 2  | 504308      | 53  | 0      | 0          |   |   |   |
| 3  | 504309      | 18  | 1      | 0          |   |   |   |
| 4  | 504310      | 52  | 1      | 1          |   |   |   |
| 5  | 504311      | 29  | 0      | 1          |   |   |   |
| 6  | 504312      | 21  | 1      | 1          |   |   |   |
| 7  | 504313      | 55  | 0      | 0          |   |   |   |
| 8  | 504314      | 17  | 1      | 0          |   |   |   |
| 9  | 504315      | 30  | 1      | 0          |   |   |   |
| 10 | 504316      | 51  | 0      | 1          |   |   |   |
| 11 | 504317      | 63  | 1      | 0          |   |   |   |
| 12 | 504318      | 26  | 0      | 0          |   |   |   |
| 13 | 504319      | 42  | 1      | 0          |   |   |   |
| 14 | 504320      | 40  | 0      | 0          |   |   |   |
| 15 | 504321      | 19  | 0      | 0          |   |   |   |
| 16 | 504322      | 30  | 1      | 0          |   |   |   |
| 17 | 504323      | 60  | 0      | 0          |   |   |   |
| 18 | 504324      | 22  | 0      | 0          |   |   |   |
| 19 | 504325      | 39  | 1      | 1          |   |   |   |
| 20 | 504326      | 21  | 1      | 0          |   |   |   |
| 21 | 504327      | 20  | 1      | 0          |   |   |   |
| 22 | 504328      | 54  | 0      | 0          |   |   |   |
| 23 | 504329      | 53  | 0      | 1          |   |   |   |
| 24 | 504330      | 22  | 1      | 0          |   |   |   |
| 25 | 504331      | 32  | 1      | 0          |   |   |   |
| 26 | 504332      | 36  | 1      | 0          |   |   |   |
| 27 | 504333      | 24  | 1      | 0          |   |   |   |
| 28 | 504334      | 57  | 1      | 0          |   |   |   |
| 29 | 504335      | 27  | 1      | 0          |   |   |   |
| 30 | 504336      | 21  | 0      | 0          |   |   |   |
| 31 | 504337      | 52  | 1      | 0          |   |   |   |
| 32 | 504338      | 52  | 1      | 1          |   |   |   |
| 33 | 504339      | 52  | 1      | 0          |   |   |   |
| 34 | 504340      | 55  | 0      | 0          |   |   |   |

Customer\_Source.csv

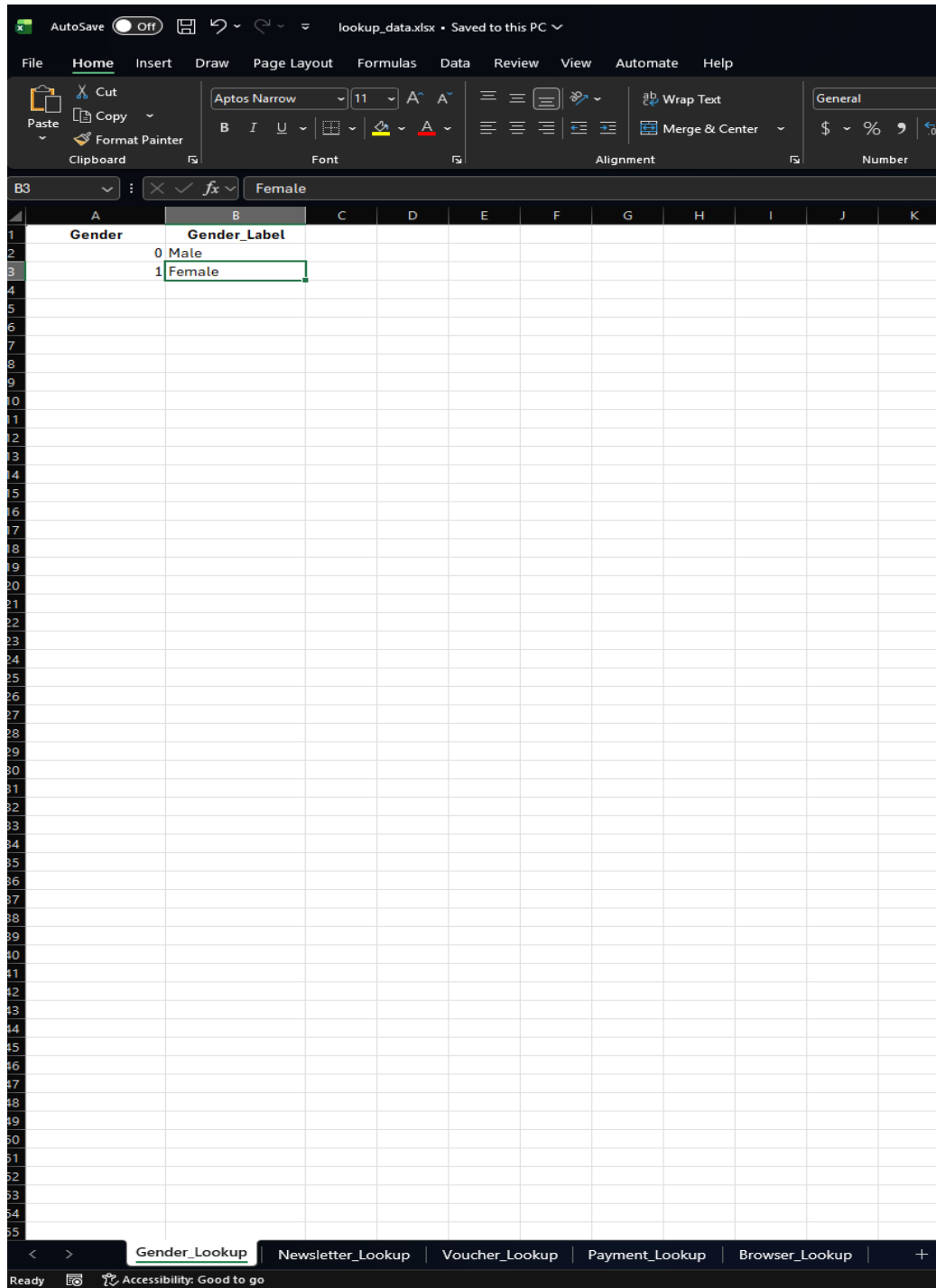
The screenshot displays the Microsoft SQL Server Enterprise Manager interface. On the left, the Object Explorer shows the server hierarchy for 'MSI\MSSQLSERVER2 (SQL Server 17.0.1000.7 - sa)'. The 'OnlineShop\_SourceDB' database is highlighted, and its 'Tables' folder is expanded, showing a table named 'dbo.Sales\_Record\_Source'. The main pane on the right shows a SQL query window titled 'SQLQuery1.sql...DB (SA (97))\*'. The query contains the following T-SQL commands:

```
1 CREATE DATABASE OnlineShop_SourceDB;  
2 GO  
3  
4 USE OnlineShop_SourceDB;  
5 GO  
6  
7 CREATE TABLE Sales_Record_Source (  
8     Customer_id INT,  
9     Purchase_DATE DATE,  
10    Revenue_Total DECIMAL(10,2),  
11    N_Purchases INT,  
12    Purchase_VALUE DECIMAL(10,2),  
13    Pay_Method INT,  
14    Time_Spent INT,  
15    Browser INT,  
16    Voucher INT  
17 );  
18 GO
```

Below the query window, the 'Messages' pane shows the following output:

```
100 %  
No issues found  
Commands completed successfully.  
Completion time: 2026-03-25T12:49:09.1823723+05:30
```

SQL Server source table



### Lookup\_data.xlsx

- This excel file contain 5 lookup sheets. The above is to show all 5 sheets in one screenshot. The screenshots of the 5 sheets separately are given below.

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B3 : X ✓ fx Female

|    | A      | B            | C | D | E | F |
|----|--------|--------------|---|---|---|---|
| 1  | Gender | Gender_Label |   |   |   |   |
| 2  |        | 0 Male       |   |   |   |   |
| 3  |        | 1 Female     |   |   |   |   |
| 4  |        |              |   |   |   |   |
| 5  |        |              |   |   |   |   |
| 6  |        |              |   |   |   |   |
| 7  |        |              |   |   |   |   |
| 8  |        |              |   |   |   |   |
| 9  |        |              |   |   |   |   |
| 10 |        |              |   |   |   |   |
| 11 |        |              |   |   |   |   |
| 12 |        |              |   |   |   |   |
| 13 |        |              |   |   |   |   |
| 14 |        |              |   |   |   |   |
| 15 |        |              |   |   |   |   |
| 16 |        |              |   |   |   |   |
| 17 |        |              |   |   |   |   |
| 18 |        |              |   |   |   |   |
| 19 |        |              |   |   |   |   |
| 20 |        |              |   |   |   |   |
| 21 |        |              |   |   |   |   |
| 22 |        |              |   |   |   |   |
| 23 |        |              |   |   |   |   |
| 24 |        |              |   |   |   |   |
| 25 |        |              |   |   |   |   |
| 26 |        |              |   |   |   |   |
| 27 |        |              |   |   |   |   |
| 28 |        |              |   |   |   |   |
| 29 |        |              |   |   |   |   |
| 30 |        |              |   |   |   |   |
| 31 |        |              |   |   |   |   |
| 32 |        |              |   |   |   |   |
| 33 |        |              |   |   |   |   |
| 34 |        |              |   |   |   |   |
| 35 |        |              |   |   |   |   |
| 36 |        |              |   |   |   |   |
| 37 |        |              |   |   |   |   |
| 38 |        |              |   |   |   |   |
| 39 |        |              |   |   |   |   |
| 40 |        |              |   |   |   |   |
| 41 |        |              |   |   |   |   |
| 42 |        |              |   |   |   |   |
| 43 |        |              |   |   |   |   |
| 44 |        |              |   |   |   |   |
| 45 |        |              |   |   |   |   |
| 46 |        |              |   |   |   |   |
| 47 |        |              |   |   |   |   |
| 48 |        |              |   |   |   |   |
| 49 |        |              |   |   |   |   |
| 50 |        |              |   |   |   |   |
| 51 |        |              |   |   |   |   |
| 52 |        |              |   |   |   |   |
| 53 |        |              |   |   |   |   |
| 54 |        |              |   |   |   |   |
| 55 |        |              |   |   |   |   |

Gender\_Lookup Newsletter\_Lookup Voucher\_Lookup

Ready Accessibility: Good to go

Gender\_Lookup

AutoSave Off lookup\_data.xlsx

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B3 : X ✓ fx Subscribed

|    | A          | B                | C | D | E | F |
|----|------------|------------------|---|---|---|---|
| 1  | Newsletter | Newsletter_Label |   |   |   |   |
| 2  |            | 0 Not subscribed |   |   |   |   |
| 3  |            | 1 Subscribed     |   |   |   |   |
| 4  |            |                  |   |   |   |   |
| 5  |            |                  |   |   |   |   |
| 6  |            |                  |   |   |   |   |
| 7  |            |                  |   |   |   |   |
| 8  |            |                  |   |   |   |   |
| 9  |            |                  |   |   |   |   |
| 10 |            |                  |   |   |   |   |
| 11 |            |                  |   |   |   |   |
| 12 |            |                  |   |   |   |   |
| 13 |            |                  |   |   |   |   |
| 14 |            |                  |   |   |   |   |
| 15 |            |                  |   |   |   |   |
| 16 |            |                  |   |   |   |   |
| 17 |            |                  |   |   |   |   |
| 18 |            |                  |   |   |   |   |
| 19 |            |                  |   |   |   |   |
| 20 |            |                  |   |   |   |   |
| 21 |            |                  |   |   |   |   |
| 22 |            |                  |   |   |   |   |
| 23 |            |                  |   |   |   |   |
| 24 |            |                  |   |   |   |   |
| 25 |            |                  |   |   |   |   |
| 26 |            |                  |   |   |   |   |
| 27 |            |                  |   |   |   |   |
| 28 |            |                  |   |   |   |   |
| 29 |            |                  |   |   |   |   |
| 30 |            |                  |   |   |   |   |
| 31 |            |                  |   |   |   |   |
| 32 |            |                  |   |   |   |   |
| 33 |            |                  |   |   |   |   |
| 34 |            |                  |   |   |   |   |
| 35 |            |                  |   |   |   |   |
| 36 |            |                  |   |   |   |   |
| 37 |            |                  |   |   |   |   |
| 38 |            |                  |   |   |   |   |
| 39 |            |                  |   |   |   |   |
| 40 |            |                  |   |   |   |   |
| 41 |            |                  |   |   |   |   |
| 42 |            |                  |   |   |   |   |
| 43 |            |                  |   |   |   |   |
| 44 |            |                  |   |   |   |   |
| 45 |            |                  |   |   |   |   |
| 46 |            |                  |   |   |   |   |
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| 48 |            |                  |   |   |   |   |
| 49 |            |                  |   |   |   |   |
| 50 |            |                  |   |   |   |   |
| 51 |            |                  |   |   |   |   |
| 52 |            |                  |   |   |   |   |
| 53 |            |                  |   |   |   |   |
| 54 |            |                  |   |   |   |   |
| 55 |            |                  |   |   |   |   |

Gender\_Lookup Newsletter\_Lookup Voucher\_Lookup

Ready Accessibility: Good to go

Newsletter\_Lookup



AutoSave Off | lookup\_data.xlsx

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Clipboard: Paste, Cut, Copy, Format Painter

Font: Aptos Narrow, 11, Bold, Italic, Underline, Color, Background Color

Alignment: B3, Used

|    | A       | B             | C | D | E | F |
|----|---------|---------------|---|---|---|---|
| 1  | Voucher | Voucher_Label |   |   |   |   |
| 2  |         | 0 Not used    |   |   |   |   |
| 3  |         | 1 Used        |   |   |   |   |
| 4  |         |               |   |   |   |   |
| 5  |         |               |   |   |   |   |
| 6  |         |               |   |   |   |   |
| 7  |         |               |   |   |   |   |
| 8  |         |               |   |   |   |   |
| 9  |         |               |   |   |   |   |
| 10 |         |               |   |   |   |   |
| 11 |         |               |   |   |   |   |
| 12 |         |               |   |   |   |   |
| 13 |         |               |   |   |   |   |
| 14 |         |               |   |   |   |   |
| 15 |         |               |   |   |   |   |
| 16 |         |               |   |   |   |   |
| 17 |         |               |   |   |   |   |
| 18 |         |               |   |   |   |   |
| 19 |         |               |   |   |   |   |
| 20 |         |               |   |   |   |   |
| 21 |         |               |   |   |   |   |
| 22 |         |               |   |   |   |   |
| 23 |         |               |   |   |   |   |
| 24 |         |               |   |   |   |   |
| 25 |         |               |   |   |   |   |
| 26 |         |               |   |   |   |   |
| 27 |         |               |   |   |   |   |
| 28 |         |               |   |   |   |   |
| 29 |         |               |   |   |   |   |
| 30 |         |               |   |   |   |   |
| 31 |         |               |   |   |   |   |
| 32 |         |               |   |   |   |   |
| 33 |         |               |   |   |   |   |
| 34 |         |               |   |   |   |   |
| 35 |         |               |   |   |   |   |
| 36 |         |               |   |   |   |   |
| 37 |         |               |   |   |   |   |
| 38 |         |               |   |   |   |   |
| 39 |         |               |   |   |   |   |
| 40 |         |               |   |   |   |   |
| 41 |         |               |   |   |   |   |
| 42 |         |               |   |   |   |   |
| 43 |         |               |   |   |   |   |
| 44 |         |               |   |   |   |   |
| 45 |         |               |   |   |   |   |
| 46 |         |               |   |   |   |   |
| 47 |         |               |   |   |   |   |
| 48 |         |               |   |   |   |   |
| 49 |         |               |   |   |   |   |
| 50 |         |               |   |   |   |   |
| 51 |         |               |   |   |   |   |
| 52 |         |               |   |   |   |   |
| 53 |         |               |   |   |   |   |
| 54 |         |               |   |   |   |   |
| 55 |         |               |   |   |   |   |

Gender\_Lookup | Newsletter\_Lookup | Voucher\_Lookup | Paym

Voucher\_Lookup

AutoSave Off | lookup\_data.xlsx

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Clipboard: Paste, Cut, Copy, Format Painter

Font: Arial, 11, Bold, Italic, Underline, Color, Background Color

Alignment: B5, Other

|    | A          | B                 | C | D | E | F | G | H |
|----|------------|-------------------|---|---|---|---|---|---|
| 1  | Pay_Method | Pay_Method_Label  |   |   |   |   |   |   |
| 2  |            | 0 Digital Wallets |   |   |   |   |   |   |
| 3  |            | 1 Card            |   |   |   |   |   |   |
| 4  |            | 2 PayPal          |   |   |   |   |   |   |
| 5  |            | 3 Other           |   |   |   |   |   |   |
| 6  |            |                   |   |   |   |   |   |   |
| 7  |            |                   |   |   |   |   |   |   |
| 8  |            |                   |   |   |   |   |   |   |
| 9  |            |                   |   |   |   |   |   |   |
| 10 |            |                   |   |   |   |   |   |   |
| 11 |            |                   |   |   |   |   |   |   |
| 12 |            |                   |   |   |   |   |   |   |
| 13 |            |                   |   |   |   |   |   |   |
| 14 |            |                   |   |   |   |   |   |   |
| 15 |            |                   |   |   |   |   |   |   |
| 16 |            |                   |   |   |   |   |   |   |
| 17 |            |                   |   |   |   |   |   |   |
| 18 |            |                   |   |   |   |   |   |   |
| 19 |            |                   |   |   |   |   |   |   |
| 20 |            |                   |   |   |   |   |   |   |
| 21 |            |                   |   |   |   |   |   |   |
| 22 |            |                   |   |   |   |   |   |   |
| 23 |            |                   |   |   |   |   |   |   |
| 24 |            |                   |   |   |   |   |   |   |
| 25 |            |                   |   |   |   |   |   |   |
| 26 |            |                   |   |   |   |   |   |   |
| 27 |            |                   |   |   |   |   |   |   |
| 28 |            |                   |   |   |   |   |   |   |
| 29 |            |                   |   |   |   |   |   |   |
| 30 |            |                   |   |   |   |   |   |   |
| 31 |            |                   |   |   |   |   |   |   |
| 32 |            |                   |   |   |   |   |   |   |
| 33 |            |                   |   |   |   |   |   |   |
| 34 |            |                   |   |   |   |   |   |   |
| 35 |            |                   |   |   |   |   |   |   |
| 36 |            |                   |   |   |   |   |   |   |
| 37 |            |                   |   |   |   |   |   |   |
| 38 |            |                   |   |   |   |   |   |   |
| 39 |            |                   |   |   |   |   |   |   |
| 40 |            |                   |   |   |   |   |   |   |
| 41 |            |                   |   |   |   |   |   |   |
| 42 |            |                   |   |   |   |   |   |   |
| 43 |            |                   |   |   |   |   |   |   |
| 44 |            |                   |   |   |   |   |   |   |
| 45 |            |                   |   |   |   |   |   |   |
| 46 |            |                   |   |   |   |   |   |   |
| 47 |            |                   |   |   |   |   |   |   |
| 48 |            |                   |   |   |   |   |   |   |
| 49 |            |                   |   |   |   |   |   |   |
| 50 |            |                   |   |   |   |   |   |   |
| 51 |            |                   |   |   |   |   |   |   |
| 52 |            |                   |   |   |   |   |   |   |
| 53 |            |                   |   |   |   |   |   |   |
| 54 |            |                   |   |   |   |   |   |   |
| 55 |            |                   |   |   |   |   |   |   |

< > Gender\_Lookup | Newsletter\_Lookup | Voucher\_Lookup | Payment\_Lookup | Browse

Ready Accessibility: Good to go

Payment\_Lookup

AutoSave Off | lookup\_data.xlsx

File Home Insert Draw Page Layout Formulas Data Review View Automate Help

Paste | Cut | Copy | Format Painter | Clipboard | Font: Arial, 11, Bold, Italic, Underline, Color | Alignment: Wrap Text, Merge & Center | Number: General, \$, %

B5 | Other

|    | A       | B             | C | D | E | F | G | H | I | J |
|----|---------|---------------|---|---|---|---|---|---|---|---|
| 1  | Browser | Browser_Label |   |   |   |   |   |   |   |   |
| 2  |         | 0 Chrome      |   |   |   |   |   |   |   |   |
| 3  |         | 1 Safari      |   |   |   |   |   |   |   |   |
| 4  |         | 2 Edge        |   |   |   |   |   |   |   |   |
| 5  |         | 3 Other       |   |   |   |   |   |   |   |   |
| 6  |         |               |   |   |   |   |   |   |   |   |
| 7  |         |               |   |   |   |   |   |   |   |   |
| 8  |         |               |   |   |   |   |   |   |   |   |
| 9  |         |               |   |   |   |   |   |   |   |   |
| 10 |         |               |   |   |   |   |   |   |   |   |
| 11 |         |               |   |   |   |   |   |   |   |   |
| 12 |         |               |   |   |   |   |   |   |   |   |
| 13 |         |               |   |   |   |   |   |   |   |   |
| 14 |         |               |   |   |   |   |   |   |   |   |
| 15 |         |               |   |   |   |   |   |   |   |   |
| 16 |         |               |   |   |   |   |   |   |   |   |
| 17 |         |               |   |   |   |   |   |   |   |   |
| 18 |         |               |   |   |   |   |   |   |   |   |
| 19 |         |               |   |   |   |   |   |   |   |   |
| 20 |         |               |   |   |   |   |   |   |   |   |
| 21 |         |               |   |   |   |   |   |   |   |   |
| 22 |         |               |   |   |   |   |   |   |   |   |
| 23 |         |               |   |   |   |   |   |   |   |   |
| 24 |         |               |   |   |   |   |   |   |   |   |
| 25 |         |               |   |   |   |   |   |   |   |   |
| 26 |         |               |   |   |   |   |   |   |   |   |
| 27 |         |               |   |   |   |   |   |   |   |   |
| 28 |         |               |   |   |   |   |   |   |   |   |
| 29 |         |               |   |   |   |   |   |   |   |   |
| 30 |         |               |   |   |   |   |   |   |   |   |
| 31 |         |               |   |   |   |   |   |   |   |   |
| 32 |         |               |   |   |   |   |   |   |   |   |
| 33 |         |               |   |   |   |   |   |   |   |   |
| 34 |         |               |   |   |   |   |   |   |   |   |
| 35 |         |               |   |   |   |   |   |   |   |   |
| 36 |         |               |   |   |   |   |   |   |   |   |
| 37 |         |               |   |   |   |   |   |   |   |   |
| 38 |         |               |   |   |   |   |   |   |   |   |
| 39 |         |               |   |   |   |   |   |   |   |   |
| 40 |         |               |   |   |   |   |   |   |   |   |
| 41 |         |               |   |   |   |   |   |   |   |   |
| 42 |         |               |   |   |   |   |   |   |   |   |
| 43 |         |               |   |   |   |   |   |   |   |   |
| 44 |         |               |   |   |   |   |   |   |   |   |
| 45 |         |               |   |   |   |   |   |   |   |   |
| 46 |         |               |   |   |   |   |   |   |   |   |
| 47 |         |               |   |   |   |   |   |   |   |   |
| 48 |         |               |   |   |   |   |   |   |   |   |
| 49 |         |               |   |   |   |   |   |   |   |   |
| 50 |         |               |   |   |   |   |   |   |   |   |
| 51 |         |               |   |   |   |   |   |   |   |   |
| 52 |         |               |   |   |   |   |   |   |   |   |
| 53 |         |               |   |   |   |   |   |   |   |   |
| 54 |         |               |   |   |   |   |   |   |   |   |
| 55 |         |               |   |   |   |   |   |   |   |   |

Gender\_Lookup | Newsletter\_Lookup | Voucher\_Lookup | Payment\_Lookup | Browser\_Lookup

Ready | Accessibility: Good to go

Browser\_Lookup

### 3: Solution architecture

Using different processes, architectures, and technologies, data from multiple source types can be collected, prepared, transformed, and converted into meaningful business insights for analysis, reporting, and decision-making.

In this solution, there are **five main components** in the proposed DW & BI architecture:

1. **Source Files / Source Systems**
2. **ETL Process**
3. **Staging Database**
4. **Data Warehouse**
5. **Reporting / BI Layer**

- **Source files / source systems:**

The data comes from three prepared sources. These are a **CSV file** for customer details, an **Excel file** for lookup details, and an **SQL Server table** for sales record details.

- **ETL process:**

ETL means **Extract, Transform, and Load**. In this step, data is taken from the source files and source database, cleaned and converted into the correct format, and then loaded into the staging database. After that, it is again transformed and loaded into the final data warehouse.

- **Staging database:**

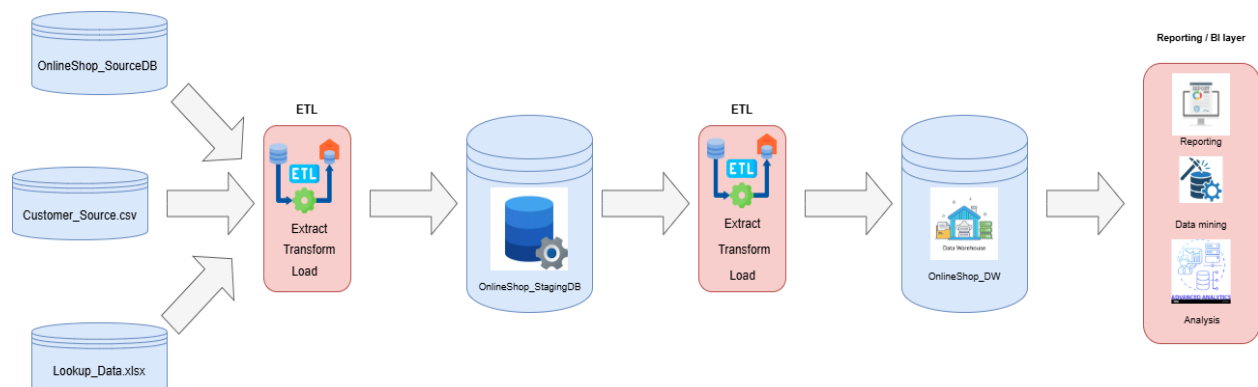
The staging database is a temporary area where data is stored before loading it into the data warehouse. It helps to organize the data and prepare it for the next step.

- **Data warehouse:**

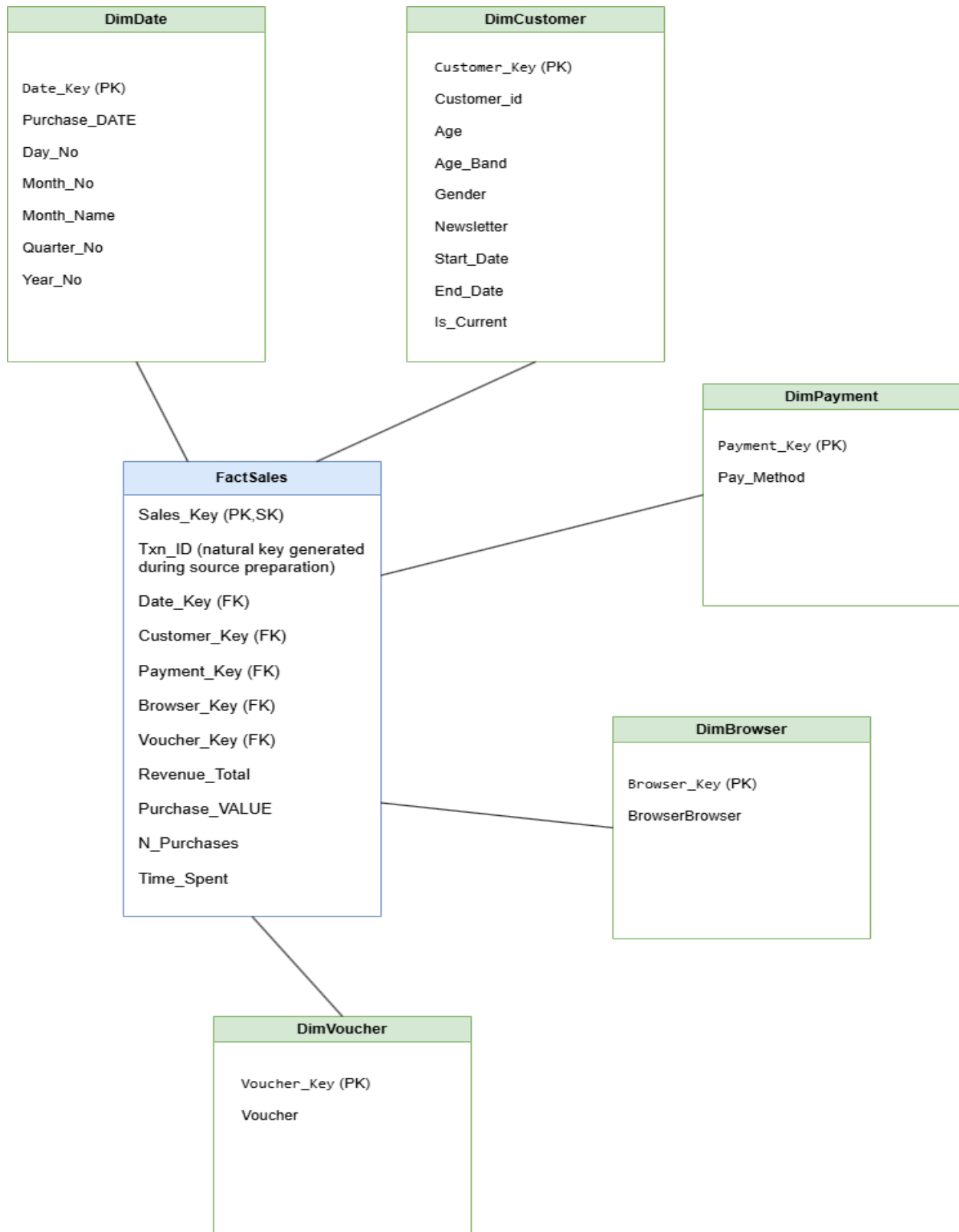
The data warehouse stores the final processed data in a structured form using fact and dimension tables. This data can then be used for analysis, reporting, and business insights.

- **Reporting layer:**

After the data is stored in the warehouse, it can be used to create reports, dashboards, and business intelligence analysis.



## 4. Data Warehouse Design & Development



## 4.1: Assumptions

- Star **schema** is used for data warehouse design.
- There is **one fact table** and **five dimensions**, including **DimDate**.
- DimCustomer is considered as the **slowly changing dimension**.
- Each source record represents **one online shop sales record**.
- A generated **Txn\_ID** is used because the original dataset does not contain a transaction identifier.
- The original flat dataset is separated into **customer**, **sales**, and **lookup** sources.
- Age\_**Band** is derived from **Age** for analysis purposes.
- DimDate is created first before the other dimensions and the fact table.
- DimCustomer, **DimPayment**, **DimBrowser**, and **DimVoucher** are loaded from the staging data.
- FactSales are loaded after all dimensions are created.

```
SQLQuery2.s...DB (SA (81))*  SQLQuery1.sq...DB (SA (97))*
1  CREATE TABLE dbo.DimDate
2  (
3      Date_Key          INT          NOT NULL PRIMARY KEY,
4      Full_Date         DATE         NOT NULL UNIQUE,
5      Day_No            TINYINT      NOT NULL,
6      Month_No          TINYINT      NOT NULL,
7      Month_Name        VARCHAR(20) NOT NULL,
8      Quarter_No        TINYINT      NOT NULL,
9      Year_No           SMALLINT     NOT NULL
10 );
11
12 CREATE TABLE dbo.DimCustomer
13 (
14     Customer_Key       INT IDENTITY(1,1) NOT NULL PRIMARY KEY,
15     Customer_id        INT            NOT NULL,
16     Age                INT            NOT NULL,
17     Age_Band           VARCHAR(20)    NOT NULL,
18     Gender             VARCHAR(20)    NOT NULL,
19     Newsletter         VARCHAR(20)    NOT NULL,
20     Start_Date         DATE           NOT NULL,
21     End_Date           DATE           NULL,
22     Is_Current         BIT            NOT NULL DEFAULT 1
23 );
```

```

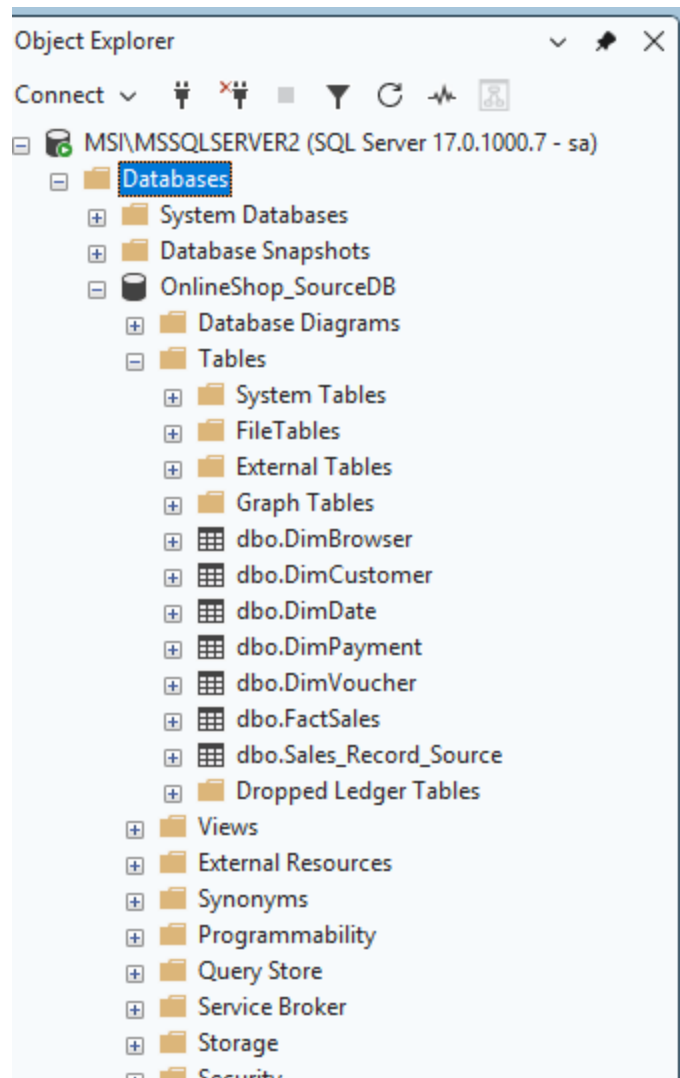
24
25  ✓ CREATE TABLE dbo.DimPayment
26  (
27      Payment_Key          INT IDENTITY(1,1) NOT NULL PRIMARY KEY,
28      Pay_Method            INT                NOT NULL,
29      Payment_Method_Label  VARCHAR(50)        NOT NULL,
30      CONSTRAINT UQ_DimPayment UNIQUE (Pay_Method)
31  );
32
33  ✓ CREATE TABLE dbo.DimBrowser
34  (
35      Browser_Key          INT IDENTITY(1,1) NOT NULL PRIMARY KEY,
36      Browser              INT                NOT NULL,
37      Browser_Label        VARCHAR(50)        NOT NULL,
38      CONSTRAINT UQ_DimBrowser UNIQUE (Browser)
39  );

```

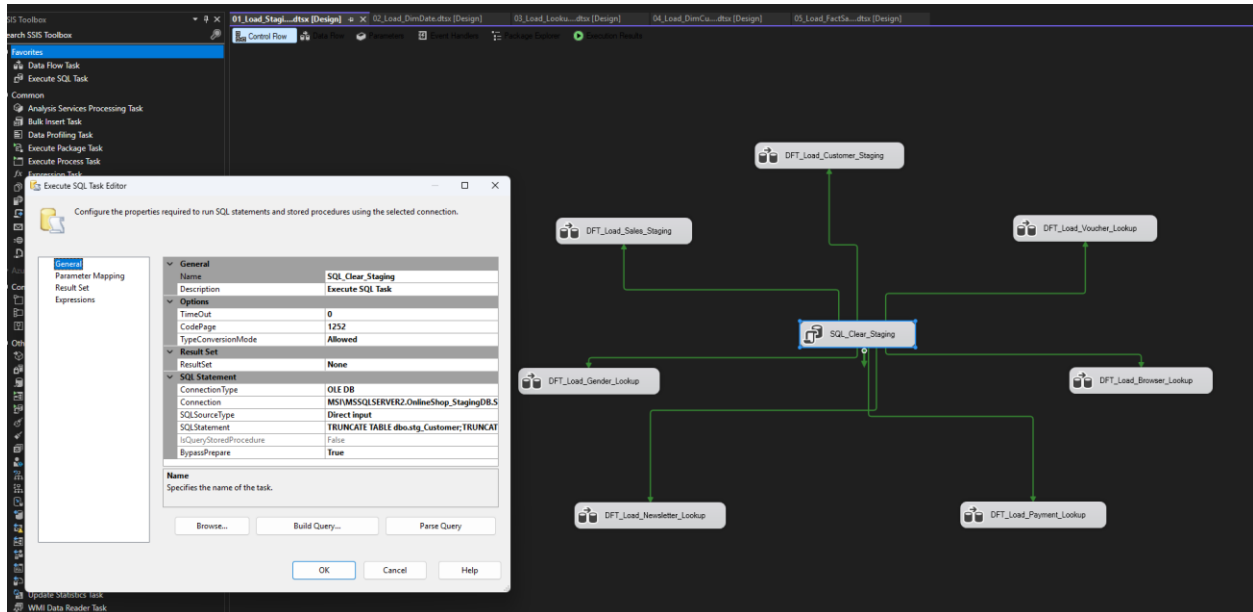
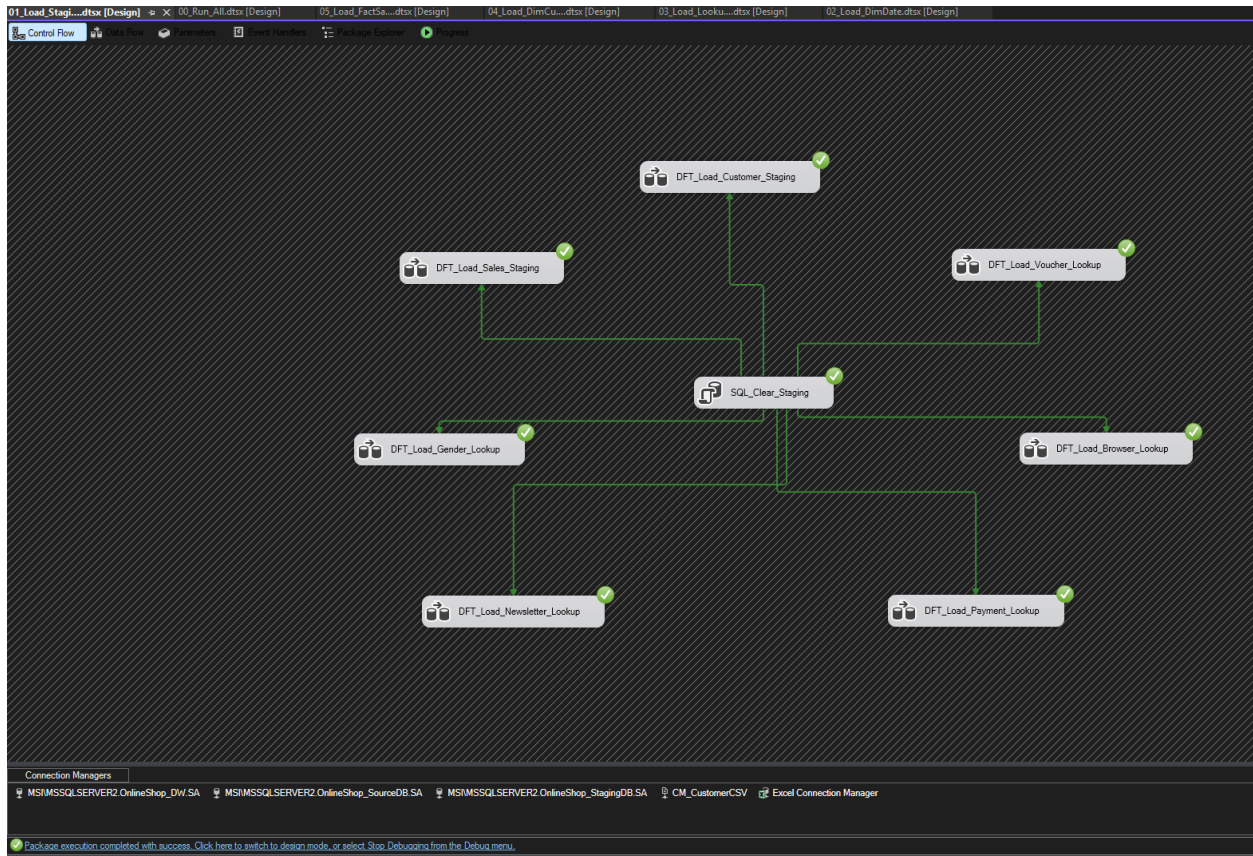
```

40
41  CREATE TABLE dbo.DimVoucher
42  (
43      Voucher_Key          INT IDENTITY(1,1) NOT NULL PRIMARY KEY,
44      Voucher              INT                NOT NULL,
45      Voucher_Label        VARCHAR(50)        NOT NULL,
46      CONSTRAINT UQ_DimVoucher UNIQUE (Voucher)
47  );
48
49  CREATE TABLE dbo.FactSales
50  (
51      Sales_Key            BIGINT IDENTITY(1,1) NOT NULL PRIMARY KEY,
52      Txn_ID               INT                NOT NULL,
53      Date_Key             INT                NOT NULL,
54      Customer_Key         INT                NOT NULL,
55      Payment_Key          INT                NOT NULL,
56      Browser_Key          INT                NOT NULL,
57      Voucher_Key          INT                NOT NULL,
58      Revenue_Total        DECIMAL(18,2)     NOT NULL,
59      Purchase_VALUE       DECIMAL(18,2)     NOT NULL,
60      N_Purchases          INT                NOT NULL,
61      Time_Spent           INT                NOT NULL,
62      accm_txn_create_time DATETIME           NULL,
63      accm_txn_complete_time DATETIME         NULL,
64      txn_process_time_hours INT              NULL,
65
66      CONSTRAINT FK_FactSales_DimDate
67          FOREIGN KEY (Date_Key) REFERENCES dbo.DimDate(Date_Key),
68
69      CONSTRAINT FK_FactSales_DimCustomer
70          FOREIGN KEY (Customer_Key) REFERENCES dbo.DimCustomer(Customer_Key),
71
72      CONSTRAINT FK_FactSales_DimPayment
73          FOREIGN KEY (Payment_Key) REFERENCES dbo.DimPayment(Payment_Key),
74
75      CONSTRAINT FK_FactSales_DimBrowser
76          FOREIGN KEY (Browser_Key) REFERENCES dbo.DimBrowser(Browser_Key),
77
78      CONSTRAINT FK_FactSales_DimVoucher
79          FOREIGN KEY (Voucher_Key) REFERENCES dbo.DimVoucher(Voucher_Key)
80  );

```



## 5: ETL Development





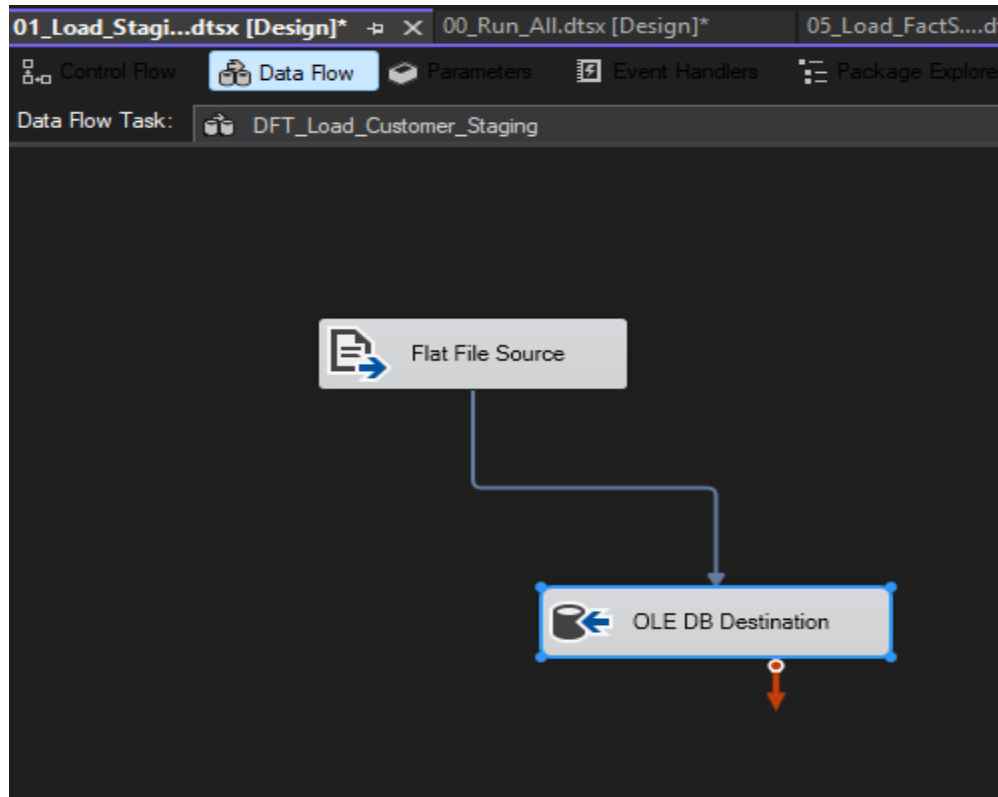
First, all required data was extracted from the source database tables and external files, then loaded into a separate staging database named OnlineShop\_StagingDB using SQL Server Integration Services (SSIS). Before loading, the staging tables were cleared, and then customer, sales, and lookup data such as gender, browser, voucher, newsletter, and payment details were populated as shown below.

The screenshot displays the SQL Server Enterprise Manager interface. The Object Explorer on the left shows the database structure of 'OnlineShop\_StagingDB', including tables like 'dbo.stg\_Customer', 'dbo.stg\_Sales\_Record', and various lookup tables. The central pane shows a SQL query executed in the 'OnlineShop\_StagingDB' database. The query counts the number of rows in several staging tables. The Results pane on the right shows the output of the query, displaying the row counts for each table.

```
1 USE OnlineShop_StagingDB;
2
3 SELECT COUNT(*) AS CustomerRows FROM dbo.stg_Customer;
4 SELECT COUNT(*) AS SalesRows FROM dbo.stg_Sales_Record;
5 SELECT COUNT(*) AS GenderRows FROM dbo.stg_Gender_Lookup;
6 SELECT COUNT(*) AS NewsletterRows FROM dbo.stg_Newsletter_Lookup;
7 SELECT COUNT(*) AS PaymentRows FROM dbo.stg_Payment_Lookup;
8 SELECT COUNT(*) AS BrowserRows FROM dbo.stg_Browser_Lookup;
9 SELECT COUNT(*) AS VoucherRows FROM dbo.stg_Voucher_Lookup;
```

| Table          | Count |
|----------------|-------|
| CustomerRows   | 65796 |
| SalesRows      | 0     |
| GenderRows     | 2     |
| NewsletterRows | 2     |
| PaymentRows    | 4     |
| BrowserRows    | 4     |
| VoucherRows    | 2     |

Query executed successfully.



01\_Load\_Stagi...dtsx [Design]\* 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCu...dtsx [Design] 05\_Load\_FactSa...dtsx [Design]

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: DFT\_Load\_Sales\_Staging

```
graph TD; A[OLE DB Source] --> B[OLE DB Destination];
```

The diagram shows a Data Flow Task named 'DFT\_Load\_Sales\_Staging'. It contains an 'OLE DB Source' component connected to an 'OLE DB Destination' component. The 'OLE DB Source' component has a red arrow pointing downwards, indicating a warning or error.

OLE DB Source Editor

Configure the properties used by a data flow to obtain data from any OLE DB provider.

Connection Manager  
Columns  
Error Output

OLE DB connection manager:  
MS/MSSQLSERVER2. OnlineShop\_SourceDB.SA New...

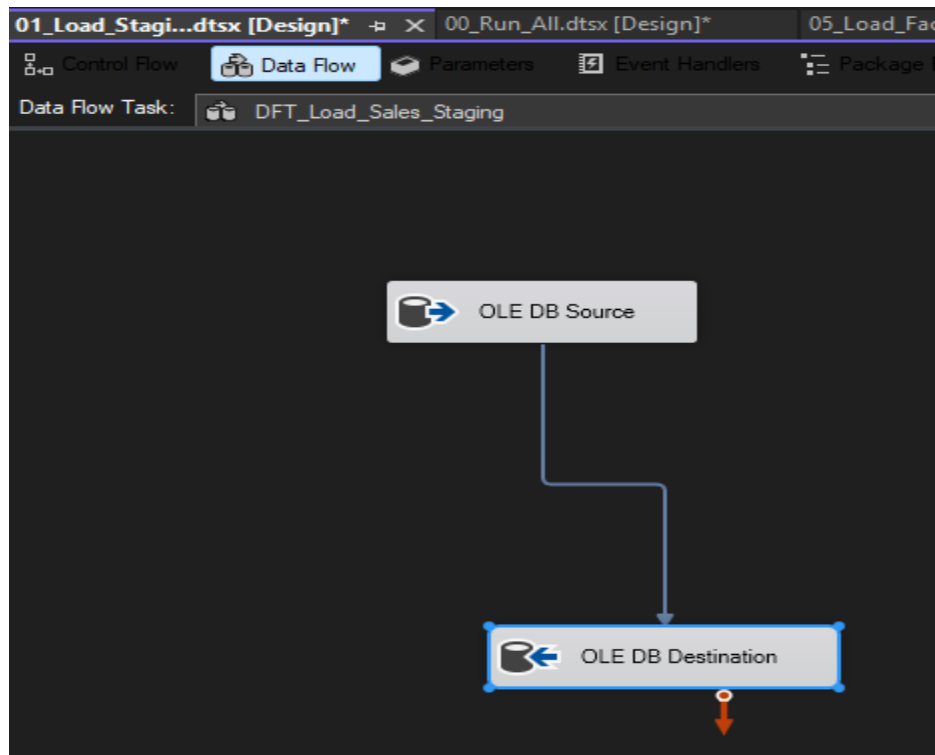
Data access mode:  
SQL command

SQL command text:  
SELECT  
Txn\_ID,  
Customer\_id,  
TRY\_CONVERT(date, Purchase\_DATE, 4) AS Purchase\_DATE,  
Revenue\_Total,  
N\_Purchases,  
Purchase\_VALUE,  
Pay\_Method,  
Time\_Spent,  
Browser,  
Voucher  
FROM dbo.Sales\_Record\_Source;

Parameters...  
Build Query...  
Browse...  
Parse Query

Preview...

OK Cancel Help



01\_Load\_Stagi...dtsx [Design]\* 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCu...dtsx [Design] 05\_Load\_FactSa...dtsx [Design]

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

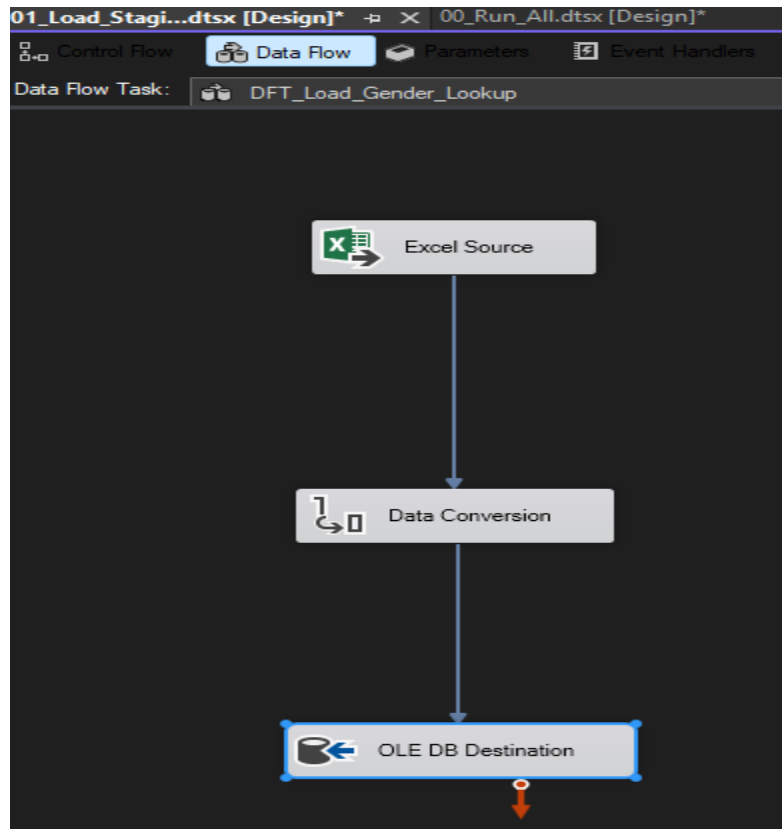
Data Flow Task: DFT\_Load\_Sales\_Staging

```
graph TD; Source[OLE DB Source] --> Destination[OLE DB Destination];
```

The screenshot shows the SSIS Data Flow Task 'DFT\_Load\_Sales\_Staging' with the 'OLE DB Destination' component highlighted. The 'OLE DB Destination Editor' dialog box is open, displaying the 'Mappings' tab. The dialog box contains a 'Connection Manager' section on the left, a 'Mappings' section in the center, and a 'Destination Column' section on the right. The 'Mappings' section shows a list of input columns from the source and a list of destination columns in the target database, with lines connecting them to indicate the mapping.

| Input Column   | Destination Column |
|----------------|--------------------|
| Txn_ID         | Txn_ID             |
| Customer_id    | Customer_id        |
| Purchase_DATE  | Purchase_DATE      |
| Revenue_Total  | Revenue_Total      |
| N_Purchases    | N_Purchases        |
| Purchase_VALUE | Purchase_VALUE     |
| Pay_Method     | Pay_Method         |
| Time_Spent     | Time_Spent         |
| Browser        | Browser            |

OK Cancel Help



Excel Source Editor

Configure the properties that enable the Data Flow task to obtain data from Excel provider.

Connection Manager  
Columns  
Error Output

Specify a connection manager, data source, or data source view for the Excel source. Then, select the mode used to access data within the source. After selecting the data access mode, select from among the additional data access options that appear.

Excel connection manager:  
Excel Connection Manager New...

Data access mode:  
Table or view

Name of the Excel sheet:  
Gender\_Lookup\$

Preview...

OK Cancel Help

Excel Source Editor

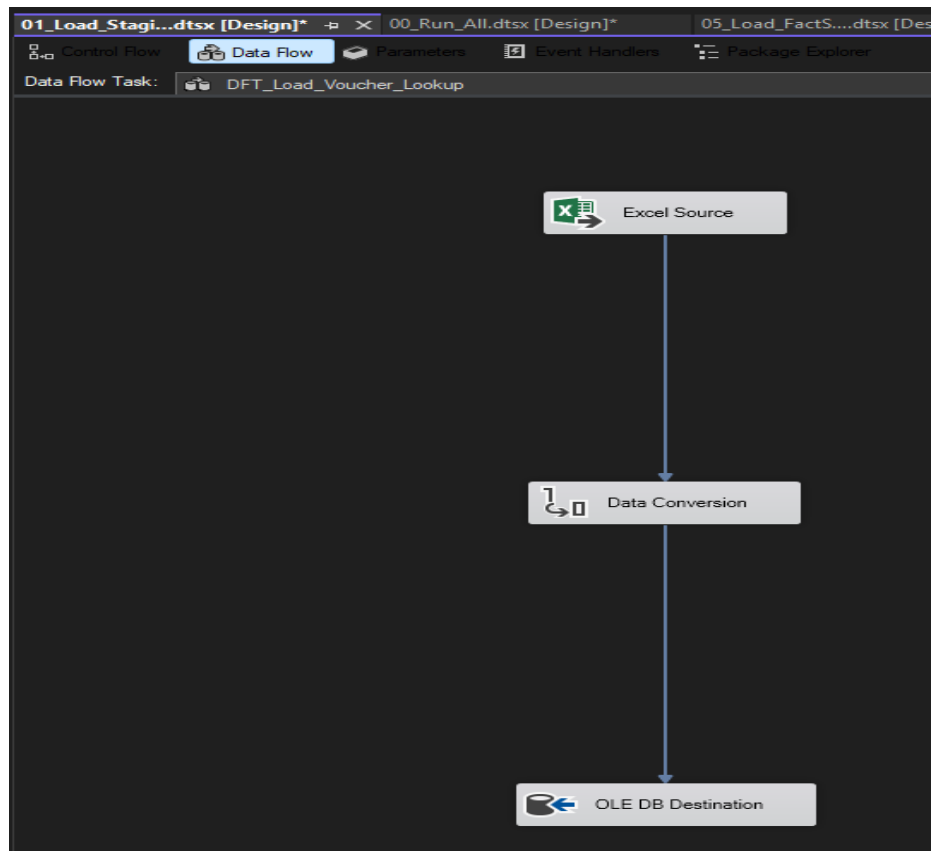
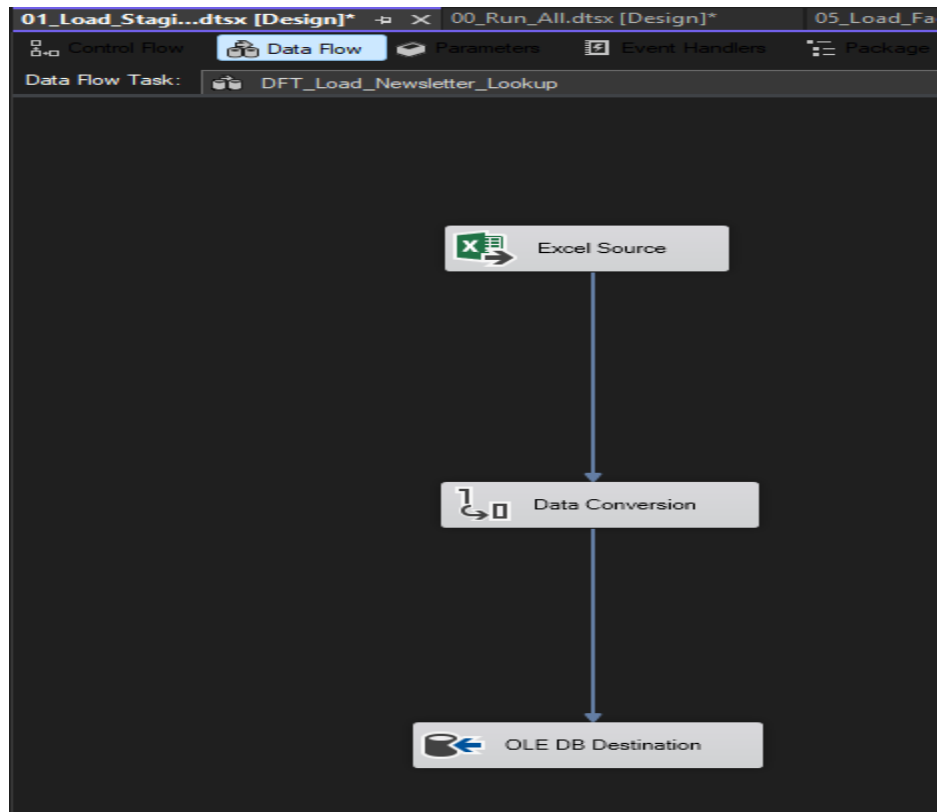
Configure the properties that enable the Data Flow task to obtain data from Excel provider.

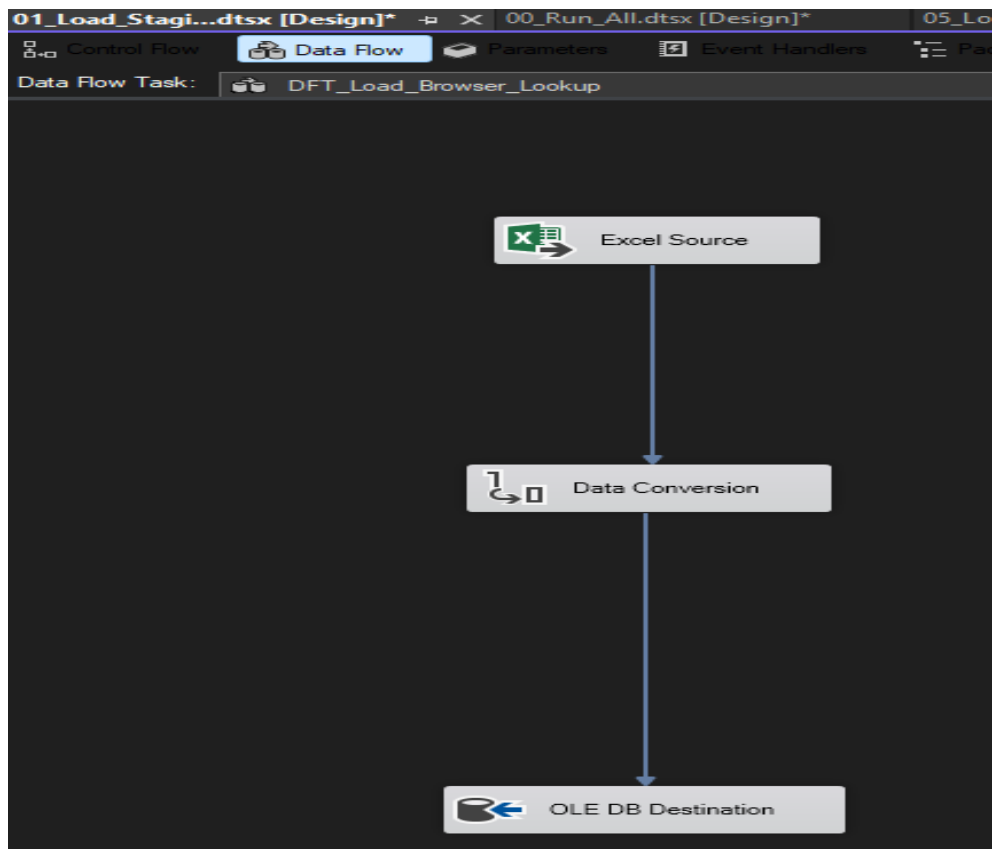
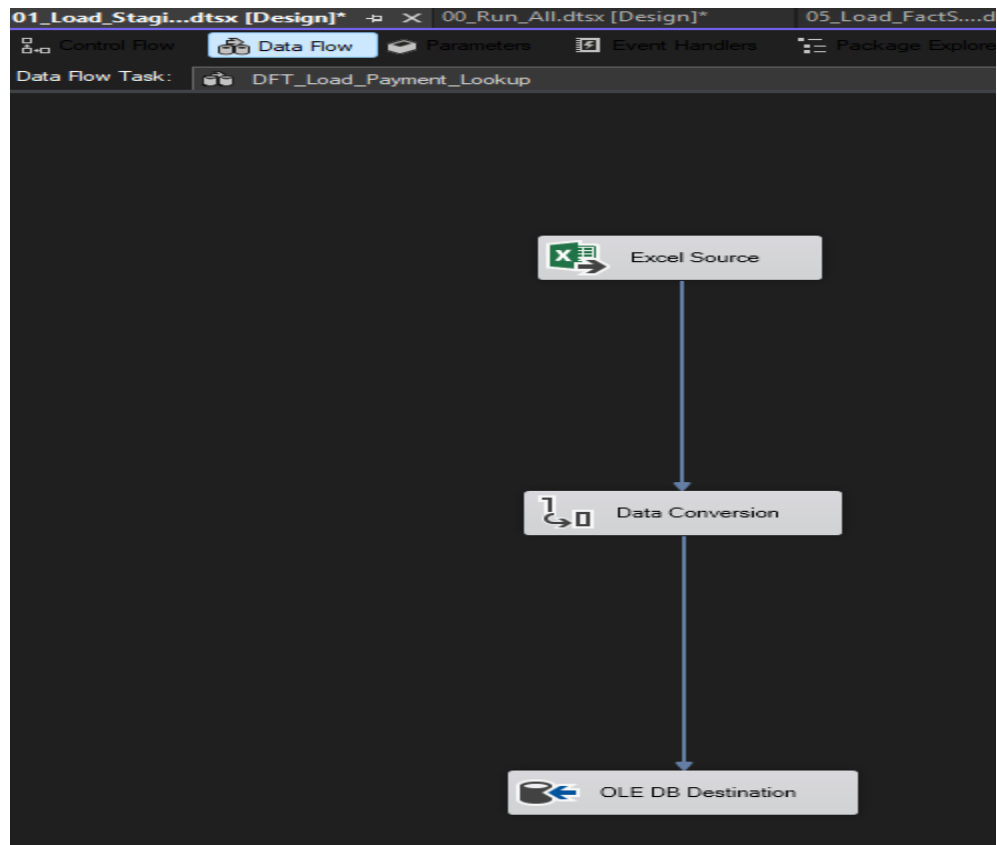
Connection Manager  
Columns  
Error Output

Available External Columns

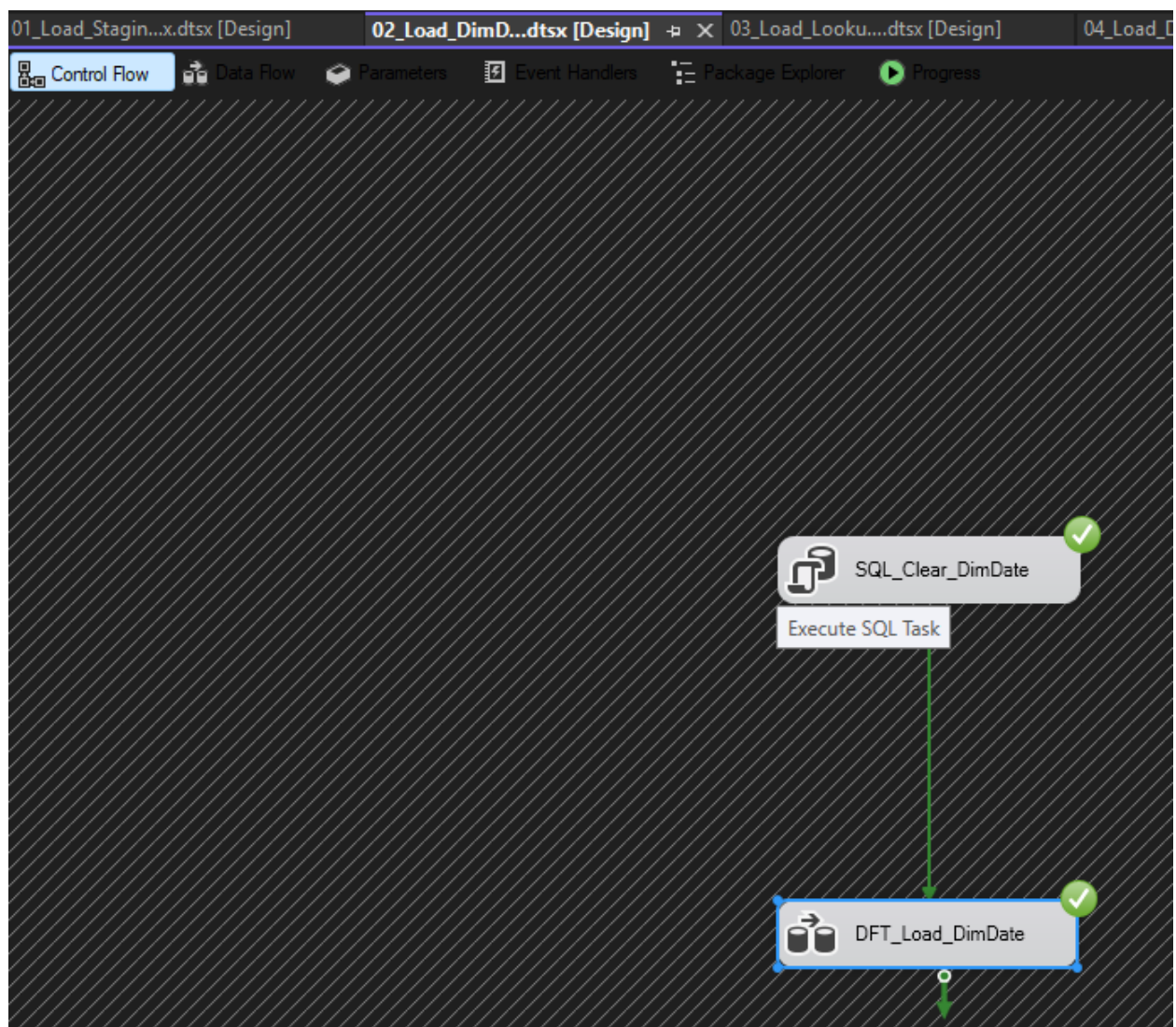
| External Column | Output Column |
|-----------------|---------------|
| Gender          | Gender        |
| Gender_Label    | Gender_Label  |

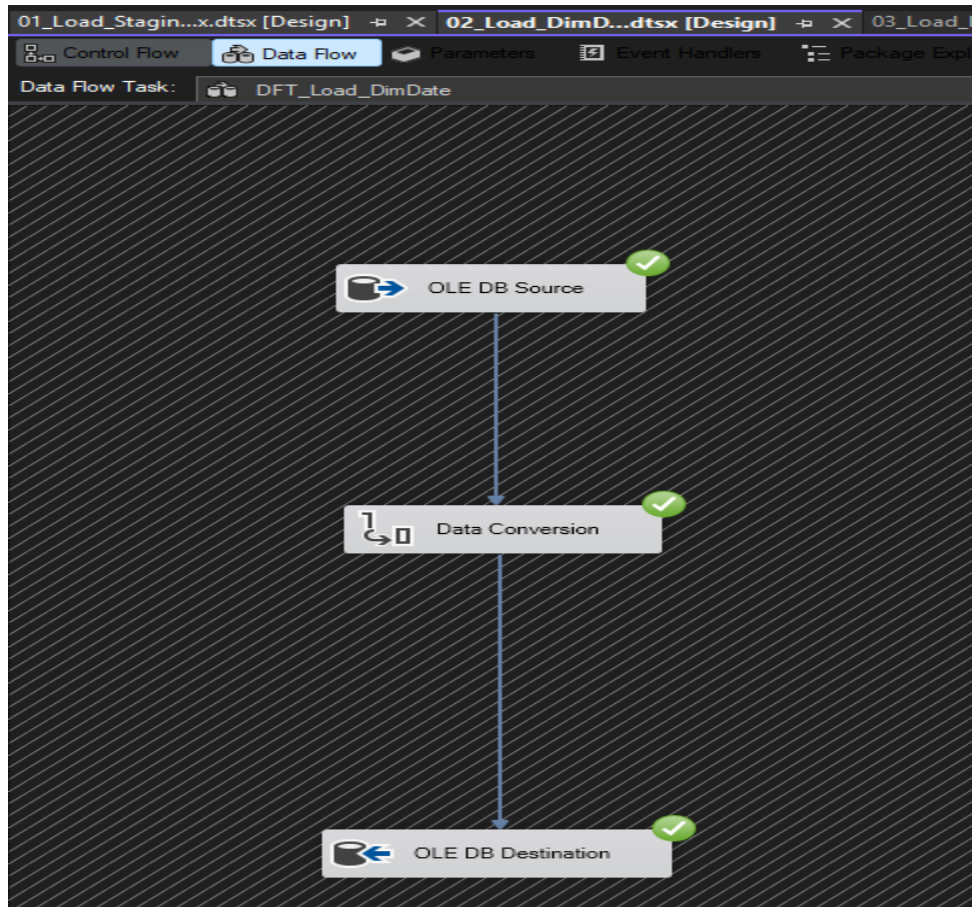
OK Cancel Help





Next, the DimDate table was loaded into the data warehouse using the 02\_Load\_DimDate.dtsx SSIS package. In this package, the existing records in the date dimension were first cleared, and then the required date-related data was loaded into the DimDate table to support time-based analysis in the warehouse.





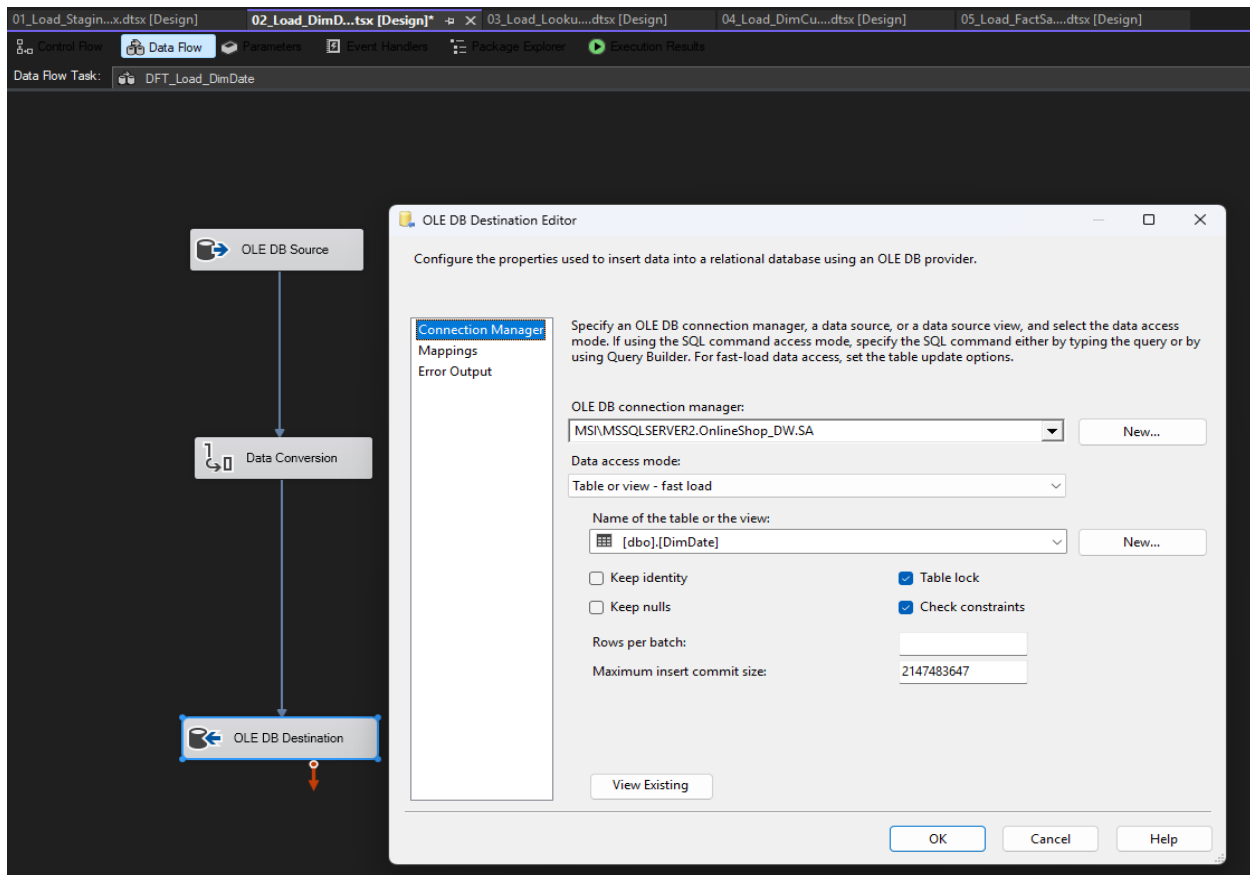
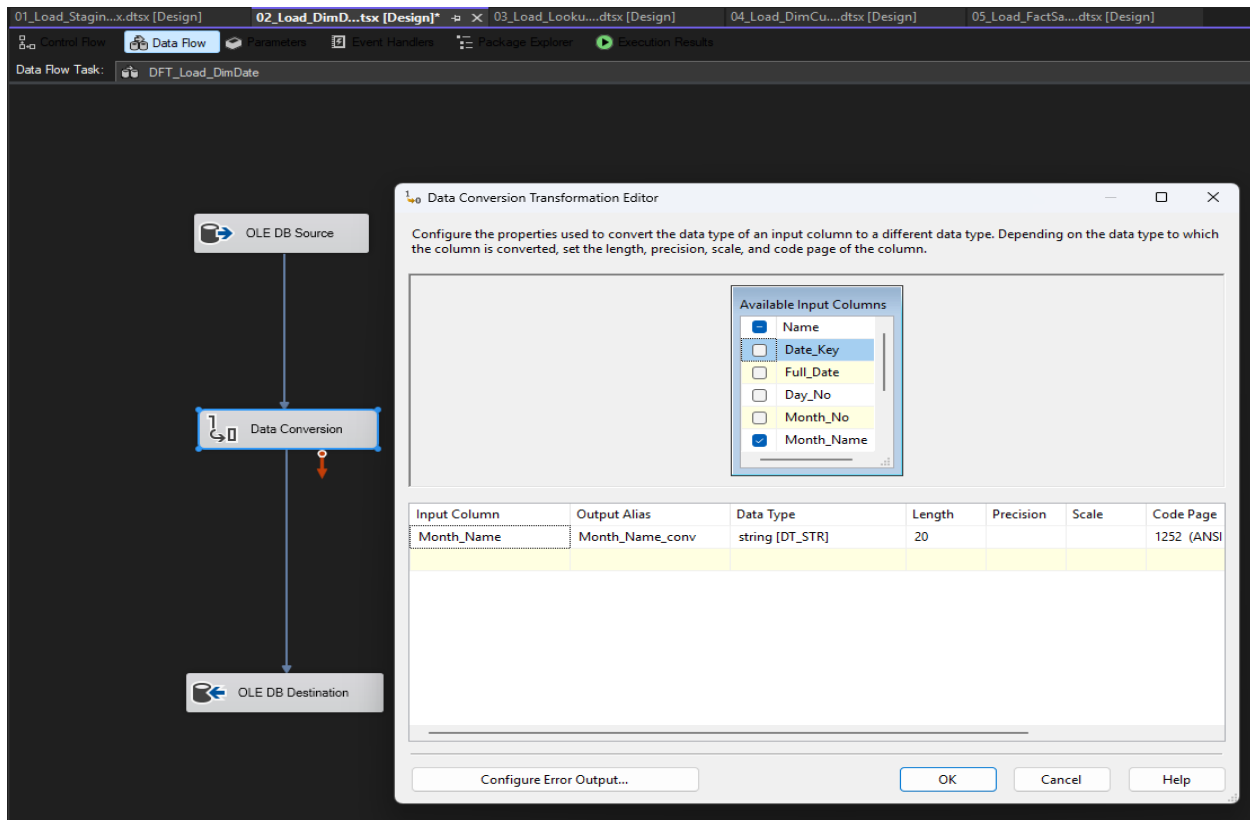
This screenshot shows the same SSIS Data Flow Task diagram as above, but with the 'OLE DB Source Editor' dialog box open. The dialog box is titled 'OLE DB Source Editor' and contains the following information:

- Connection Manager:** A list box showing 'MSI\MSSQLSERVER2.OnlineShop\_StagingDB.SA' with a 'New...' button next to it.
- Data access mode:** A dropdown menu set to 'SQL command'.
- SQL command text:** A text area containing the following SQL query:

```
SELECT DISTINCT
  CONVERT(INT, CONVERT(VARCHAR(8), Purchase_DATE, 112)) AS
  Date_Key,
  Purchase_DATE AS Full_Date,
  DAY(Purchase_DATE) AS Day_No,
  MONTH(Purchase_DATE) AS Month_No,
  DATENAME(MONTH, Purchase_DATE) AS Month_Name,
  DATEPART(QUARTER, Purchase_DATE) AS Quarter_No,
  YEAR(Purchase_DATE) AS Year_No
FROM dbo.stg_Sales_Record
WHERE Purchase_DATE IS NOT NULL;
```
- Buttons:** 'Parameters...', 'Build Query...', 'Browse...', 'Parse Query', 'Preview...', 'OK', 'Cancel', and 'Help'.

The background diagram shows the 'OLE DB Source' task selected, with a red arrow pointing to it from the dialog box.





01\_Load\_Stagin...x.dtsx [Design] 02\_Load\_DimD...tsx [Design]\* 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCu...dtsx [Design] 05\_Load\_FactSa...dtsx [Design]

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: DFT\_Load\_DimDate

```
graph TD; A[OLE DB Source] --> B[Data Conversion]; B --> C[OLE DB Destination];
```

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager  
Mappings  
Error Output

Available Input ...

| Name       |
|------------|
| Date_Key   |
| Full_Date  |
| Day_No     |
| Month_No   |
| Month_Name |
| Quarter_No |
| Year_No    |

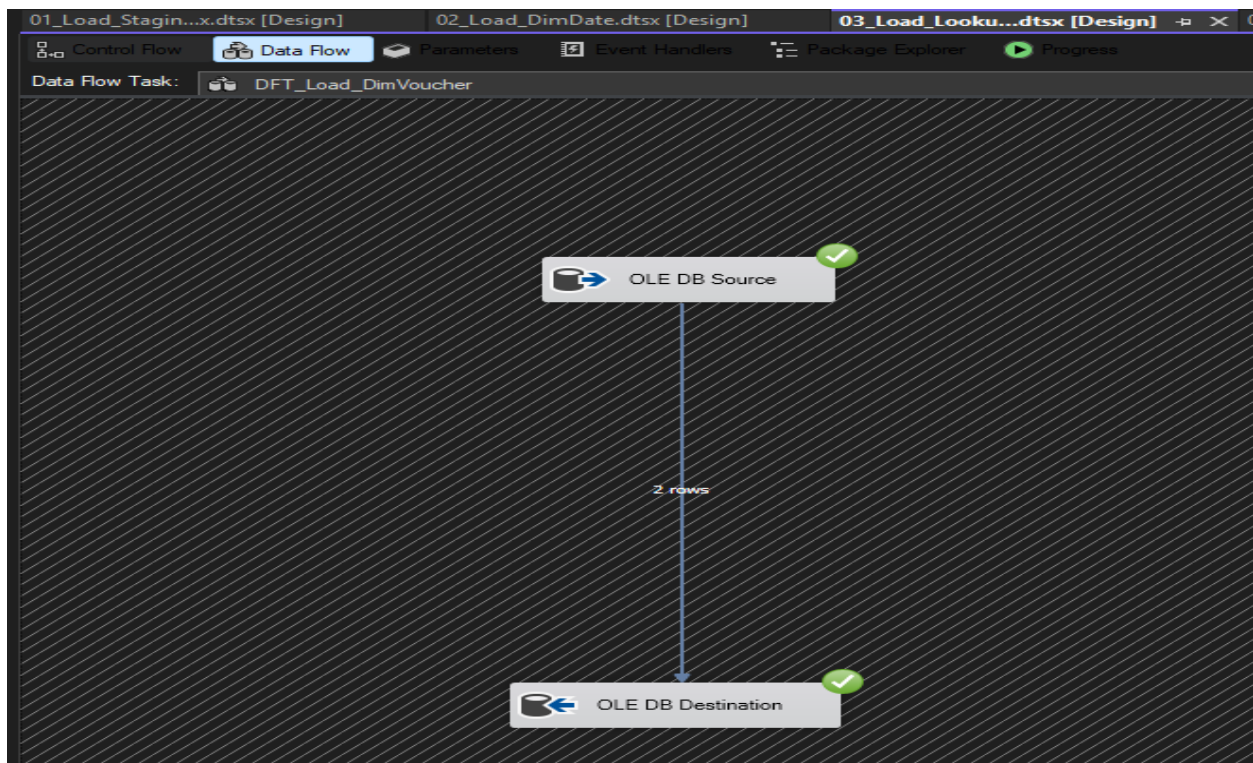
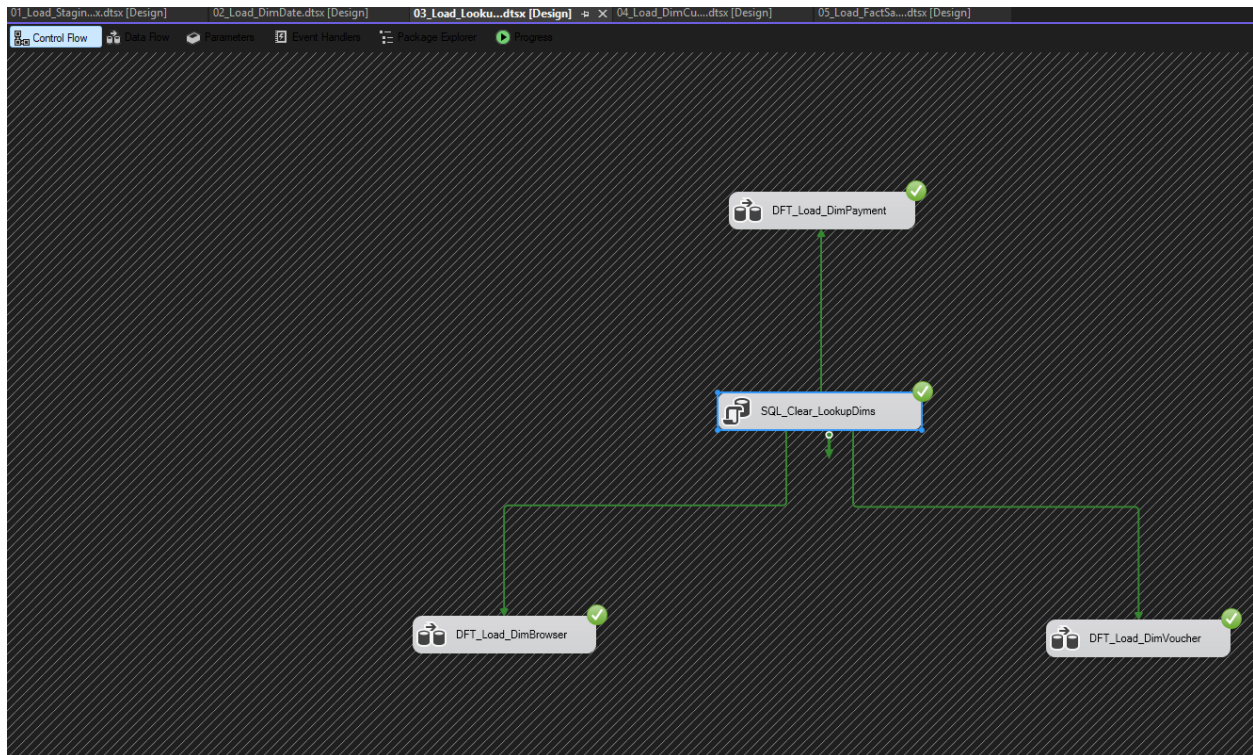
Available Destin...

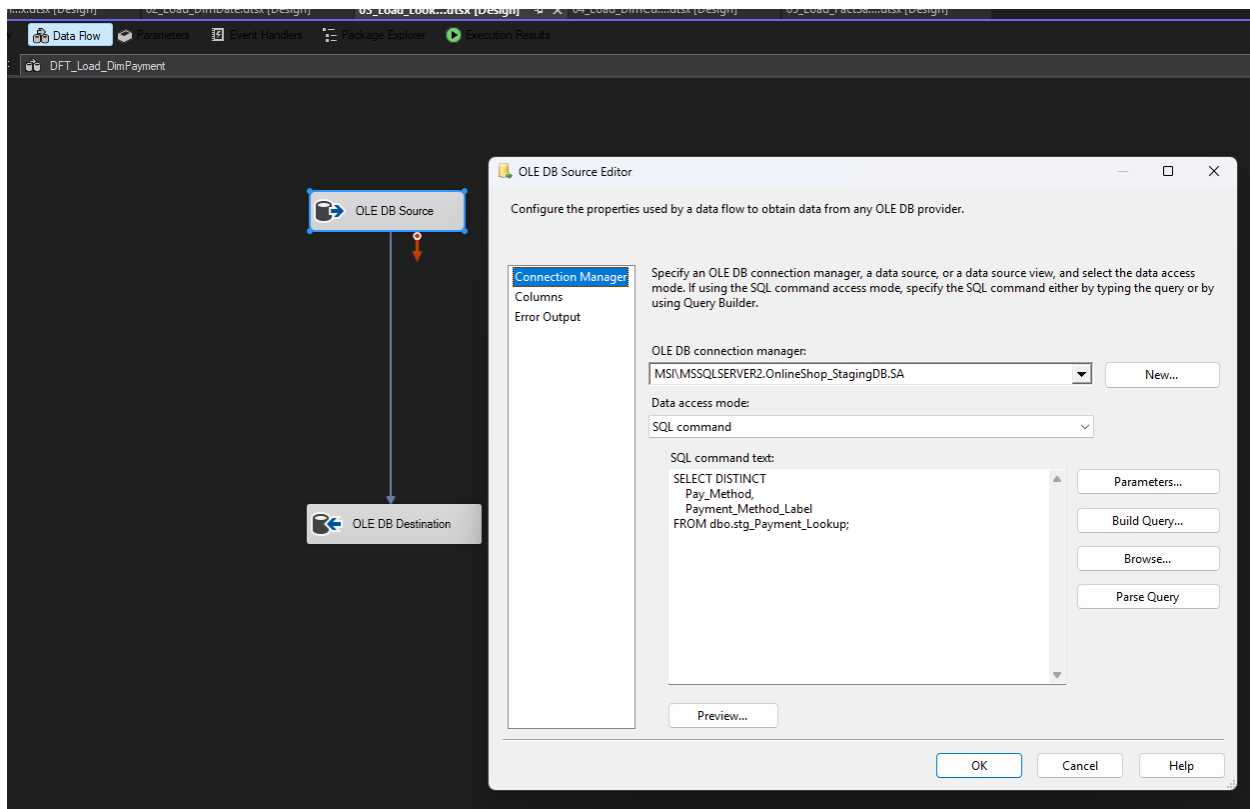
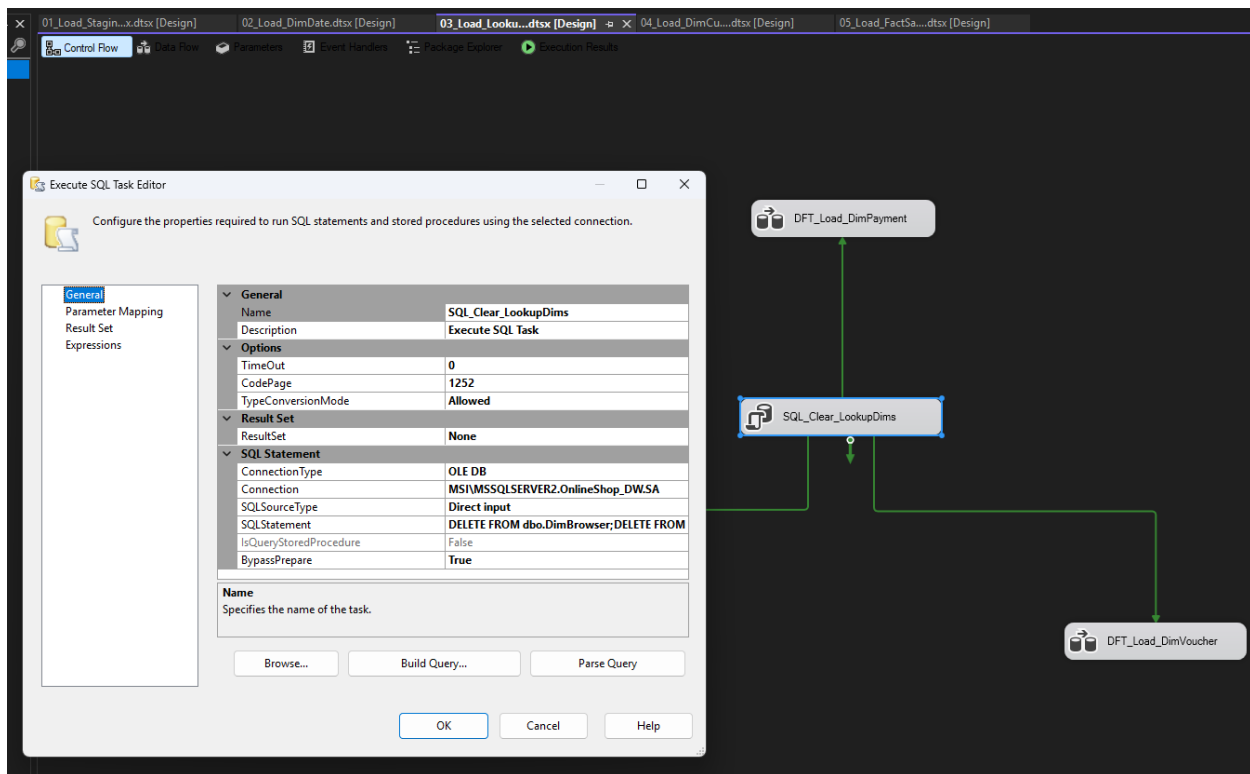
| Name       |
|------------|
| Date_Key   |
| Full_Date  |
| Day_No     |
| Month_No   |
| Month_Name |
| Quarter_No |

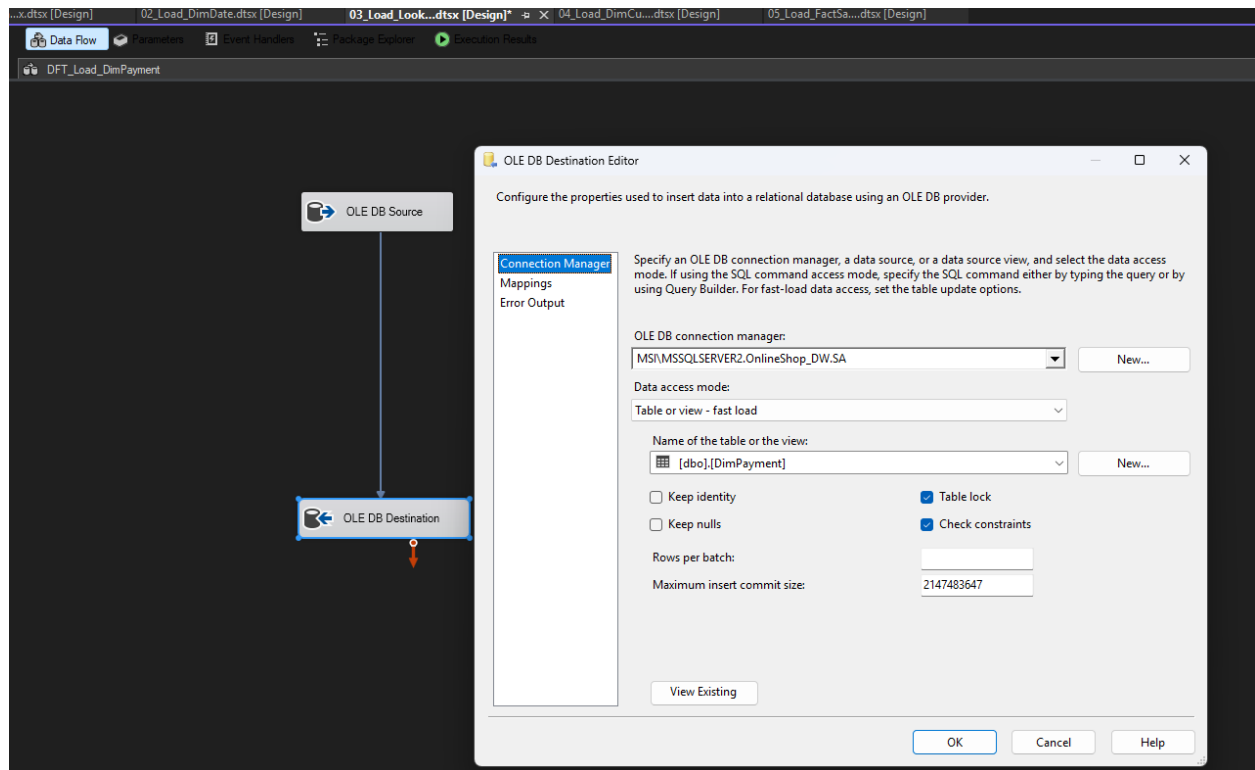
| Input Column    | Destination Column |
|-----------------|--------------------|
| Date_Key        | Date_Key           |
| Full_Date       | Full_Date          |
| Day_No          | Day_No             |
| Month_No        | Month_No           |
| Month_Name_conv | Month_Name         |
| Quarter_No      | Quarter_No         |
| Year_No         | Year_No            |

OK Cancel Help

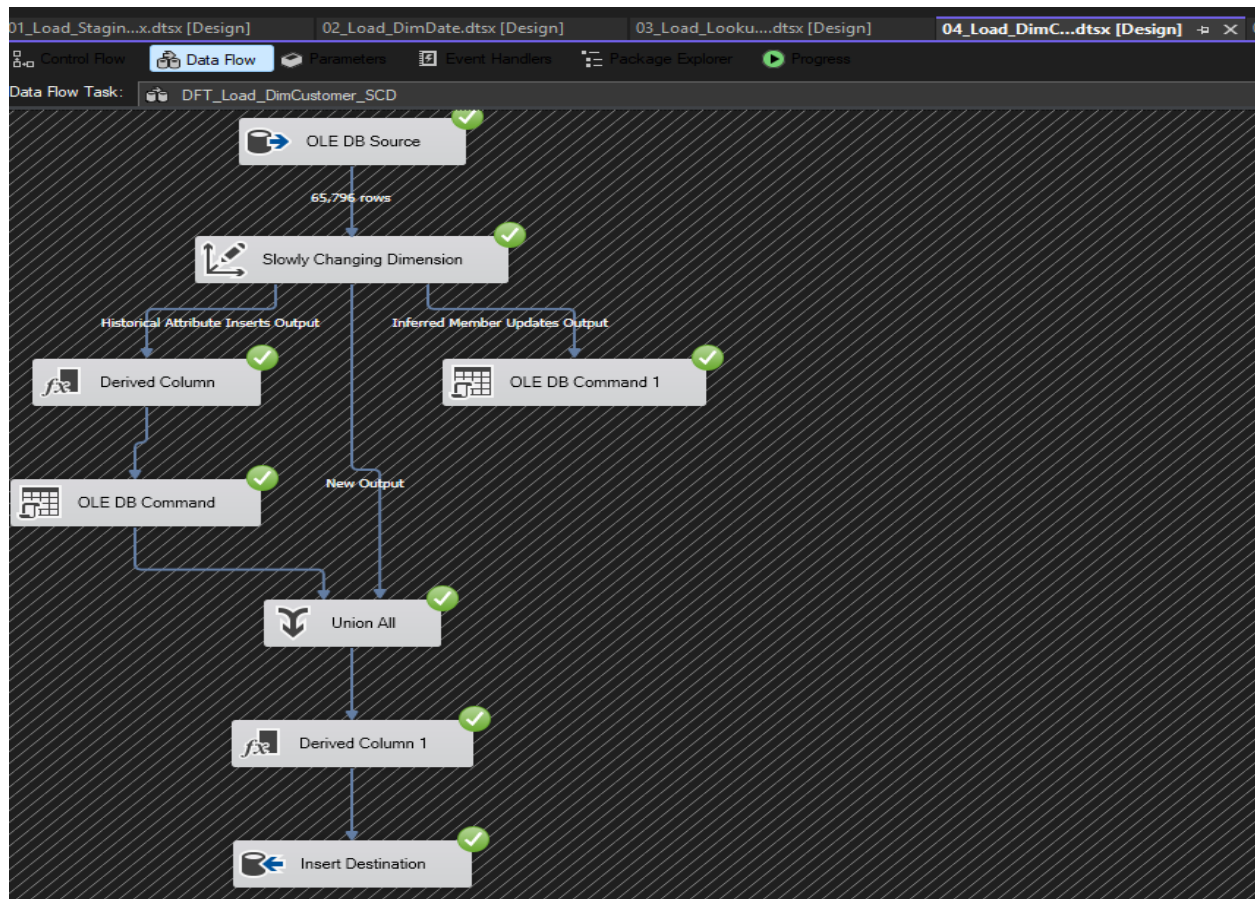
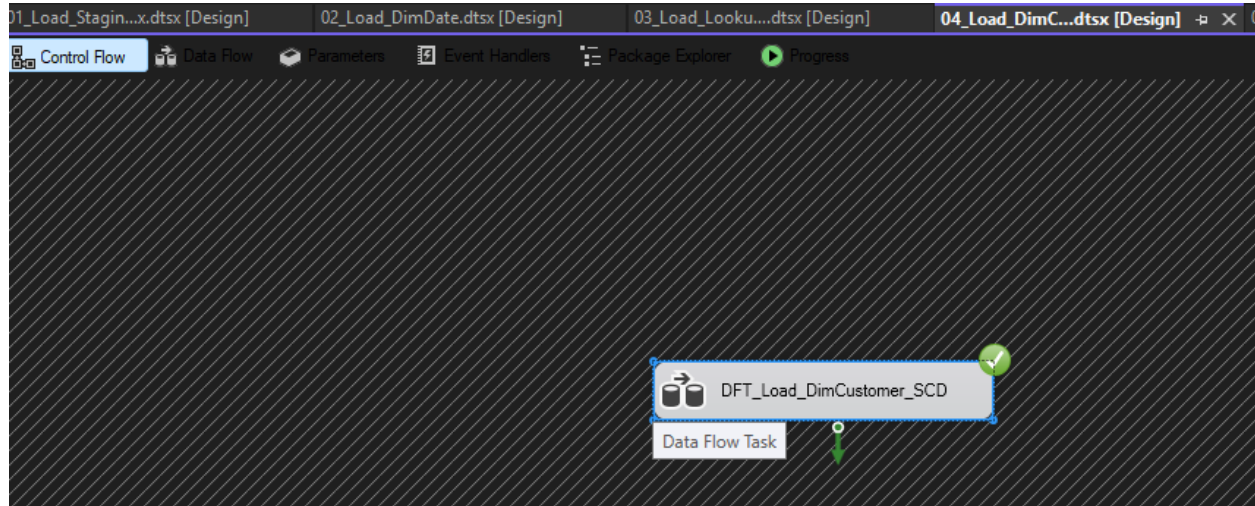
Next, the lookup dimension tables were loaded into the data warehouse using the 03\_Load\_Lookup.dtsx SSIS package. In this package, the existing records in the lookup dimension tables were first cleared, and then the payment, browser, and voucher dimension data were loaded separately to support descriptive analysis in the warehouse.







Next, the customer dimension was loaded into the data warehouse using the 04\_Load\_DimCustomer.dtsx SSIS package. In the control flow, a single data flow task was used to process the customer dimension. In the data flow, customer data was extracted from the source and passed through a Slowly Changing Dimension (SCD) transformation to identify new records, historical attribute changes, and inferred member updates before loading the results into the DimCustomer table.



01\_Load\_Stagin...x.dtsx [Design] 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCus...dtsx [Design] 05\_Load\_FactSa...dtsx [Design]

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: DFT\_Load\_DimCustomer\_SCD

OLE DB Source

Slowly Changing Dimension

Historical Attribute Inserts Output

Derived Column

OLE DB Command

Inferred Member Updates Output

OLE DB Command 1

New Output

Union All

Derived Column 1

Insert Destination

OLE DB Source Editor

Configure the properties used by a data flow to obtain data from any OLE DB provider.

Connection Manager

Columns

Error Output

OLE DB connection manager: MSN\MSSQLSERVER2.OnlineShop\_StagingDB.SA New...

Data access mode: SQL command

SQL command text:

```
SELECT
c.Customer_id,
c.Age,
CASE
WHEN c.Age <= 18 THEN '18 and below'
WHEN c.Age BETWEEN 19 AND 25 THEN '19-25'
WHEN c.Age BETWEEN 26 AND 35 THEN '26-35'
WHEN c.Age BETWEEN 36 AND 45 THEN '36-45'
WHEN c.Age BETWEEN 46 AND 55 THEN '46-55'
ELSE '56 and above'
END AS Age_Band,
g.Gender_Label AS Gender,
n.Newsletter_Label AS Newsletter
FROM dbo.stg_Customer c
```

Parameters... Build Query... Browse... Parse Query

Preview...

OK Cancel Help

01\_Load\_Stagin...x.dtsx [Design] 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCus...dtsx [Design] 05\_Load\_FactSa...dtsx [Design]

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: DFT\_Load\_DimCustomer\_SCD

OLE DB Source

Slowly Changing Dimension

Historical Attribute Inserts Output

Derived Column

OLE DB Command

Inferred Member Updates Output

OLE DB Command 1

New Output

Union All

Derived Column 1

Insert Destination

OLE DB Source Editor

Configure the properties used by a data flow to obtain data from any OLE DB provider.

Connection Manager

Columns

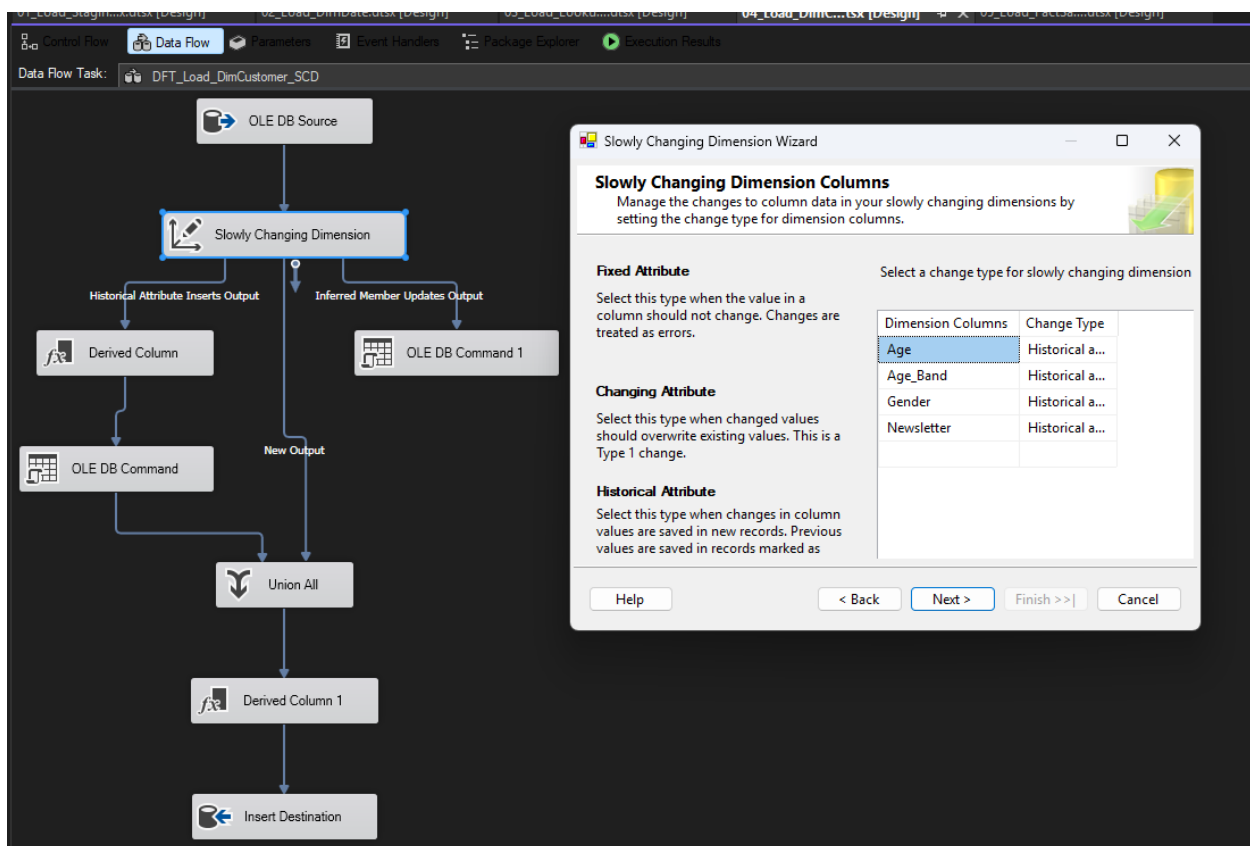
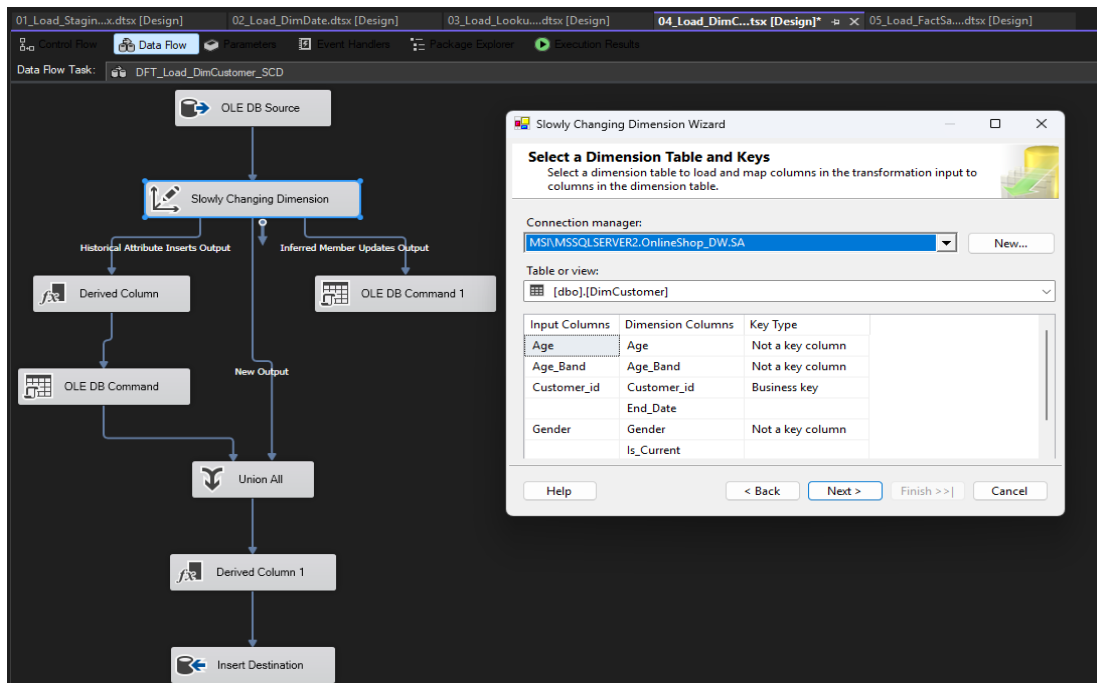
Error Output

Available External Columns

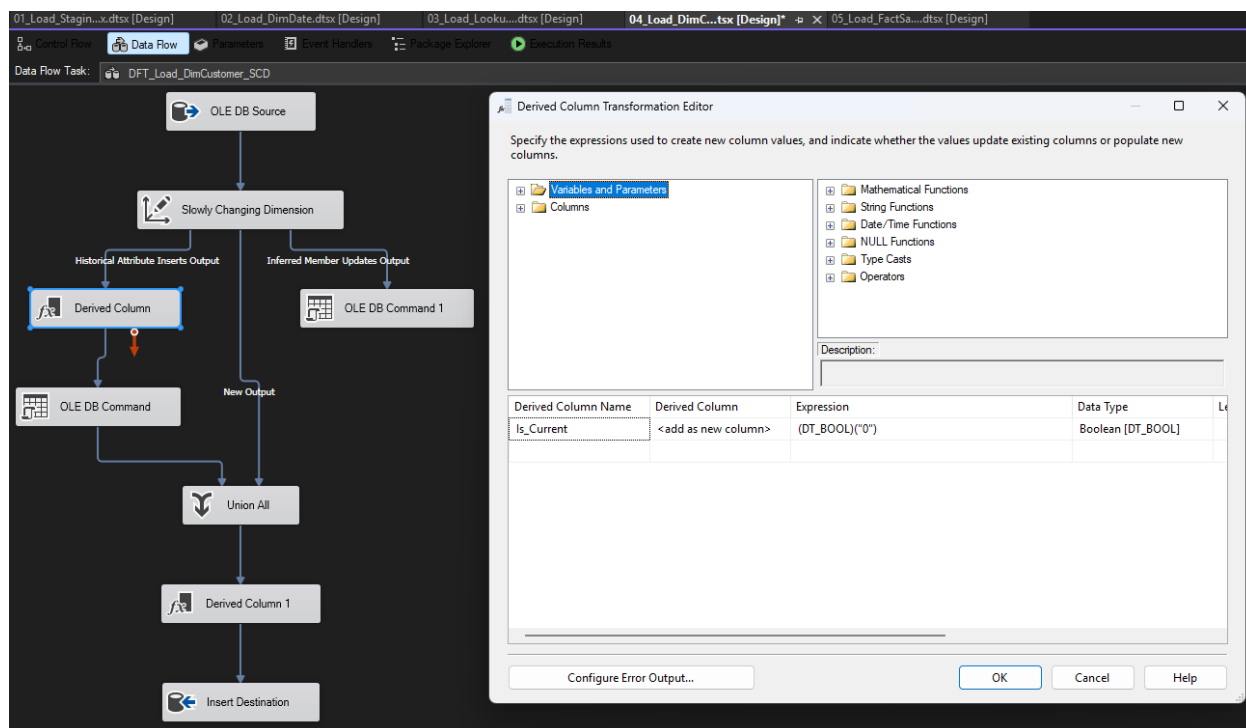
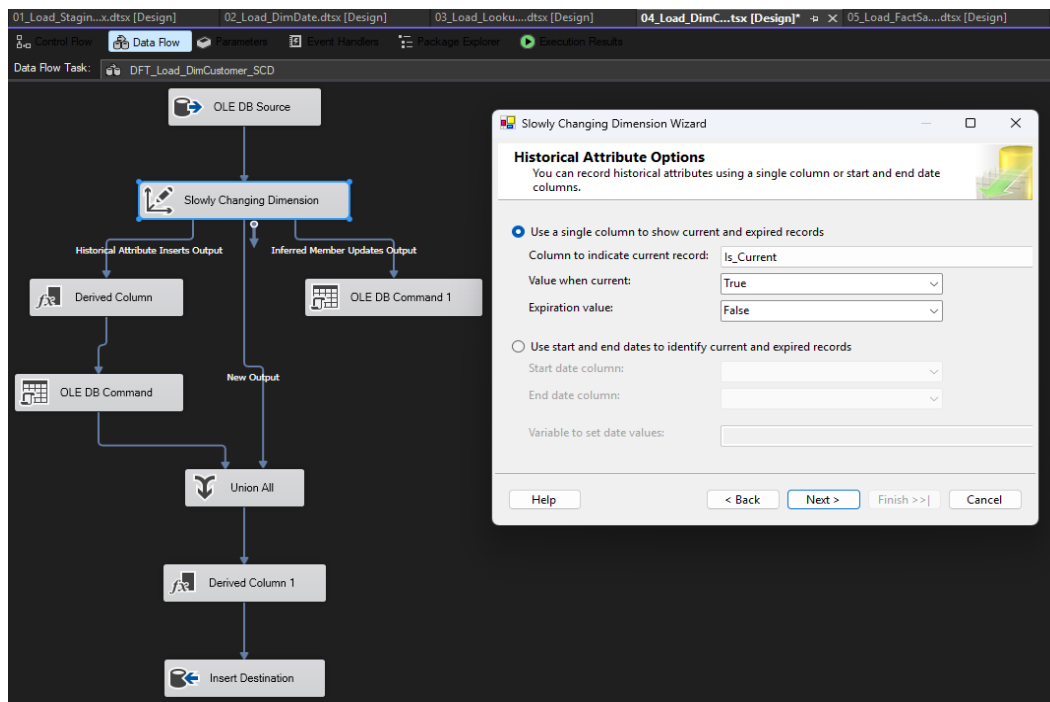
- ☒ Name
- ☒ Customer\_id
- ☒ Age
- ☒ Age\_Band
- ☒ Gender

| External Column | Output Column |
|-----------------|---------------|
| Customer_id     | Customer_id   |
| Age             | Age           |
| Age_Band        | Age_Band      |
| Gender          | Gender        |
| Newsletter      | Newsletter    |

OK Cancel Help







OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager

Mappings

Error Output

OLE DB connection manager:  
MS\MSSQLSERVER2.OnlineShop\_DW.SA

Data access mode:  
Table or view

Name of the table or the view:  
[dbo].[DimCustomer]

View Existing

OK Cancel Help

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager

Mappings

Error Output

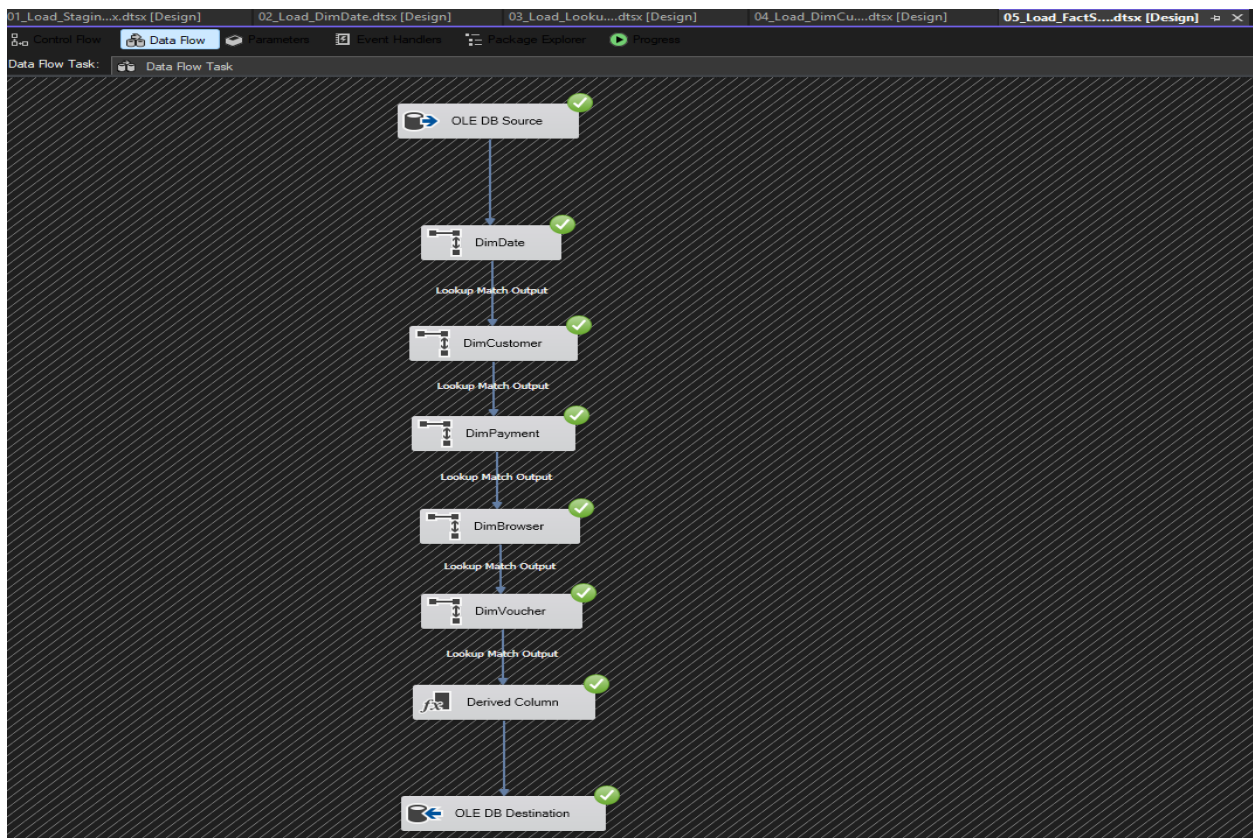
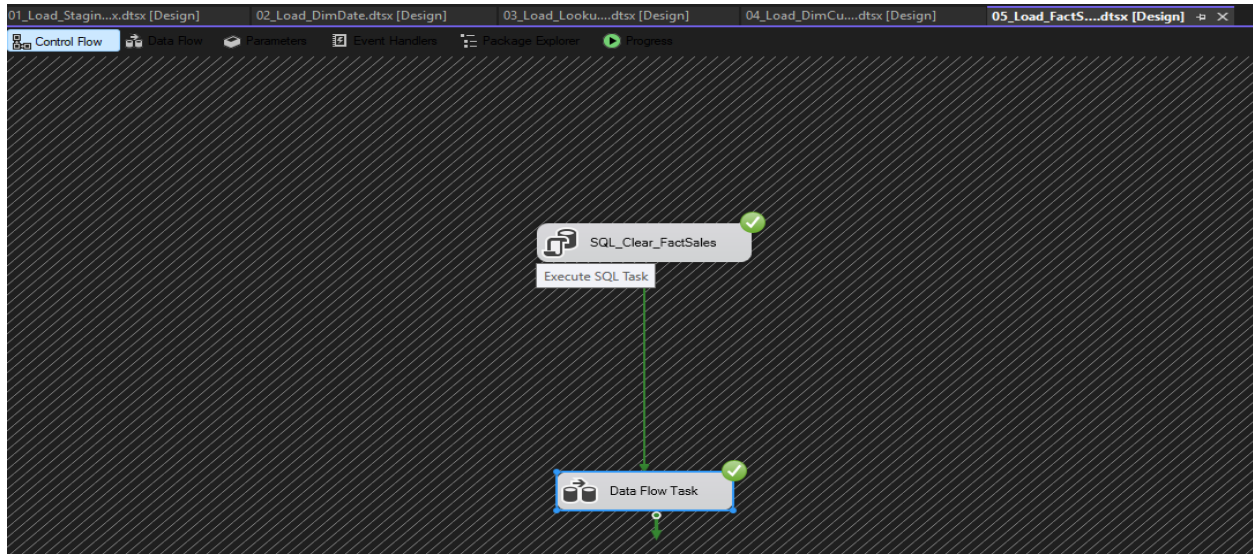
Available Input ...

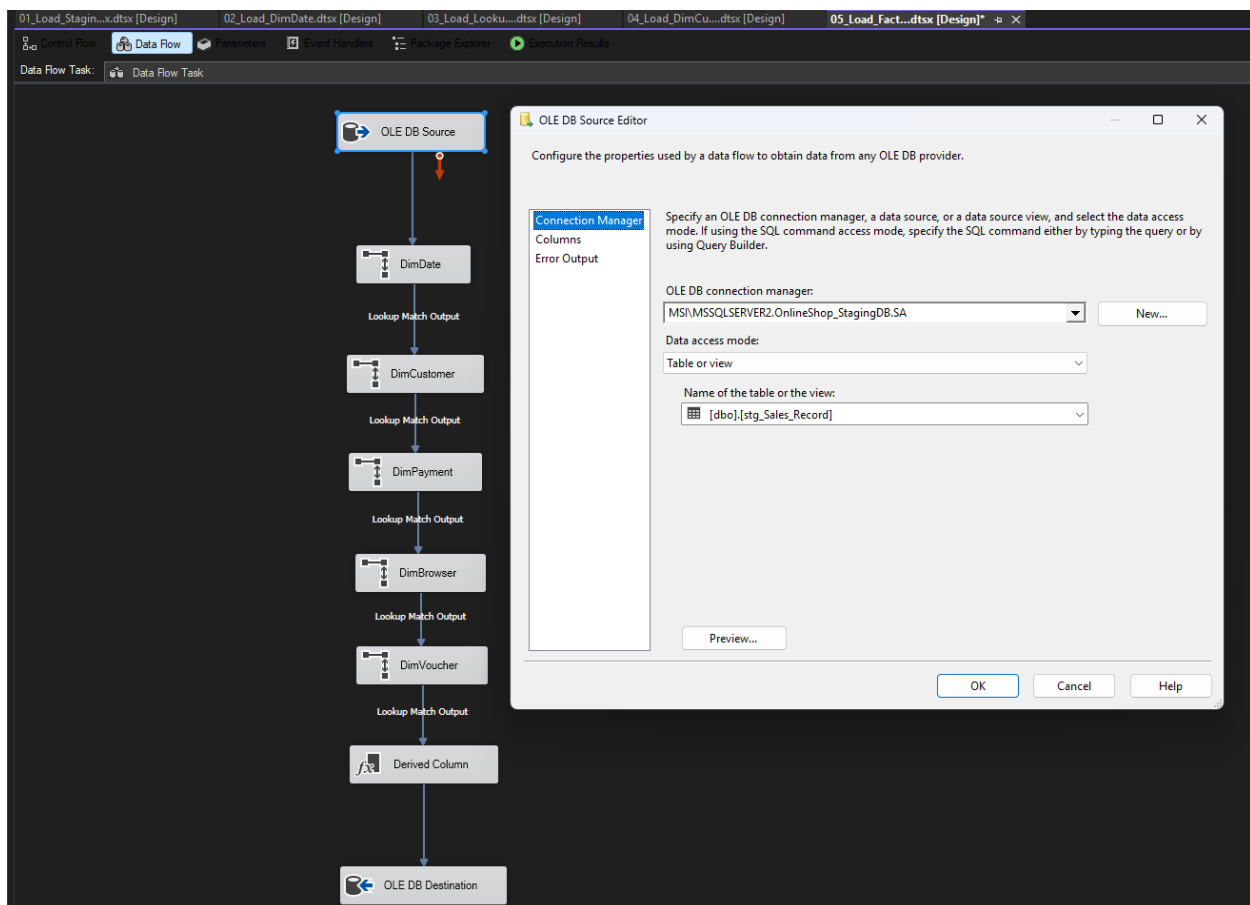
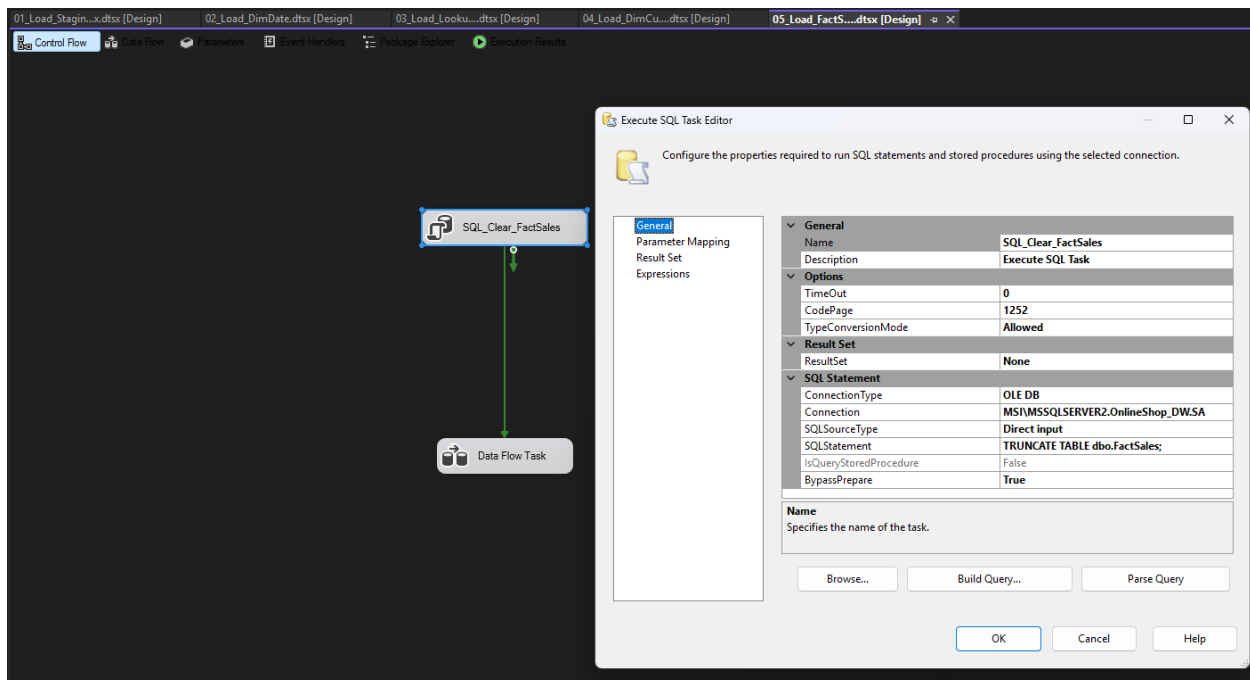
Available Destina...

| Input Column | Destination Column |
|--------------|--------------------|
| <ignore>     | Customer_Key       |
| Customer_id  | Customer_id        |
| Age          | Age                |
| Age_Band     | Age_Band           |
| Gender       | Gender             |
| Newsletter   | Newsletter         |
| <ignore>     | Start_Date         |
| <ignore>     | End_Date           |
| Is_Current   | Is_Current         |

OK Cancel Help

Finally, the fact sales table was loaded into the data warehouse using the 05\_Load\_FactSales.dtsx SSIS package. In the control flow, the existing records in the fact table were first cleared, and then the data flow task was executed to load the sales facts. Within the data flow, the source data was matched with related dimension tables such as DimDate, DimCustomer, DimPayment, DimBrowser, and DimVoucher to retrieve the required surrogate keys before inserting the final records into the fact table.





01\_Load\_Stagin...x.dtsx [Design] 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCu...dtsx [Design] 05\_Load\_Fact...dtsx [Design]\*

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: Data Flow Task

OLE DB Source Editor

Configure the properties used by a data flow to obtain data from any OLE DB provider.

Connection Manager  
Columns  
Error Output

Available External Columns

- ☒ Name
- ☒ Revenue\_Total
- ☒ N\_Purchases
- ☒ Purchase\_VALUE
- ☒ Pay\_Method
- ☒ Time\_Spent
- ☒ Browser
- ☒ Voucher

| External Column | Output Column  |
|-----------------|----------------|
| Txn_ID          | Txn_ID         |
| Customer_id     | Customer_id    |
| Purchase_DATE   | Purchase_DATE  |
| Revenue_Total   | Revenue_Total  |
| N_Purchases     | N_Purchases    |
| Purchase_VALUE  | Purchase_VALUE |
| Pay_Method      | Pay_Method     |
| Time_Spent      | Time_Spent     |
| Browser         | Browser        |
| Voucher         | Voucher        |

OK Cancel Help

01\_Load\_Stagin...x.dtsx [Design] 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCu...dtsx [Design] 05\_Load\_Fact...dtsx [Design]\*

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: Data Flow Task

Lookup Transformation Editor

This transform enables the performance of simple equi-joins between the input and a reference data set.

General  
Connection  
Columns  
Advanced  
Error Output

Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.

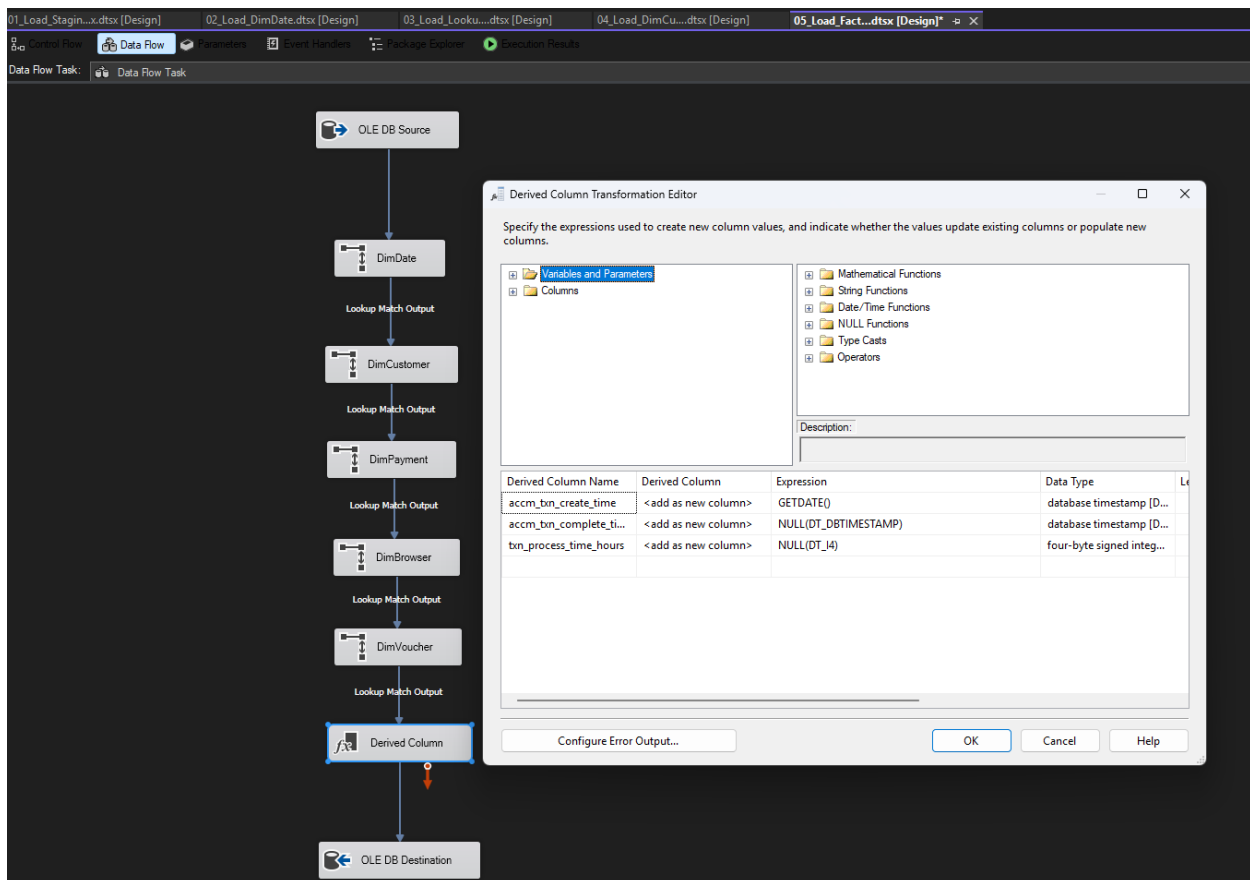
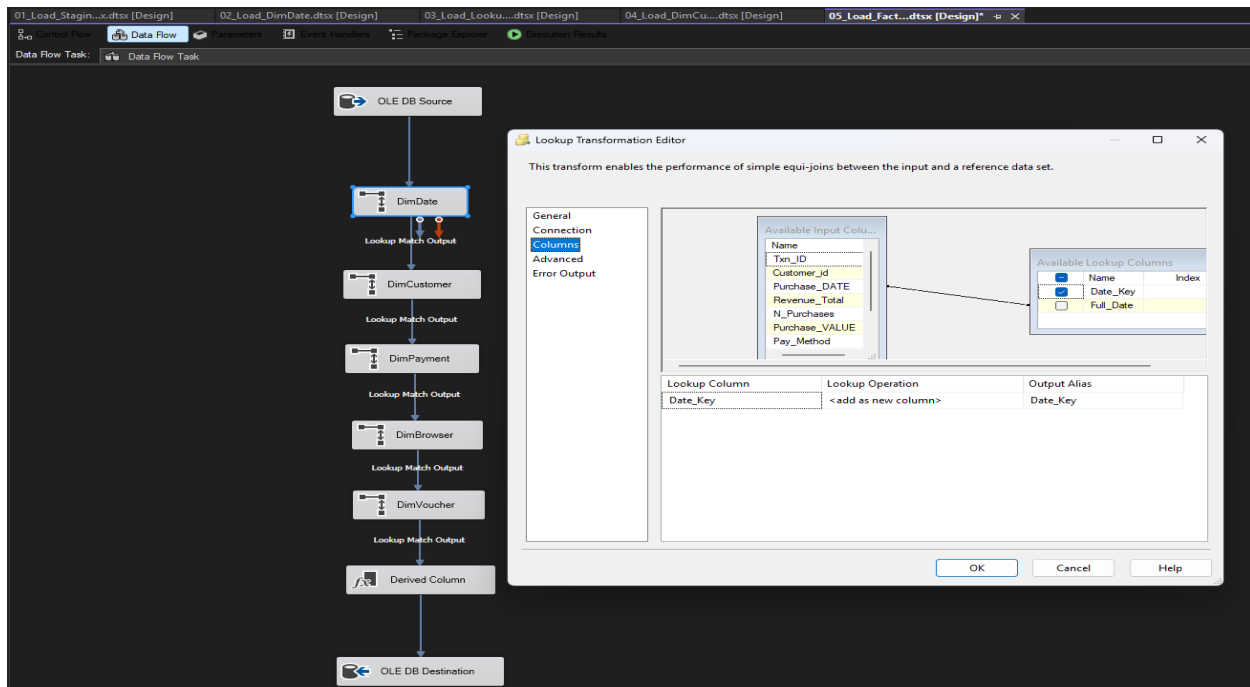
OLE DB connection manager:  
MS\MSSQLSERVER2.OnlineShop\_DW.SA New...

☐ Use a table or a view:  
New...

☒ Use results of an SQL query:  
SELECT Date\_Key, Full\_Date  
FROM dbo.DimDate;  
Build Query...  
Browse...  
Parse Query

Preview...

OK Cancel Help



OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager

OLE DB connection manager: MSN\MSSQLSERVER2.OnlineShop\_DW.SA

Data access mode: Table or view - fast load

Name of the table or the view: [dbo].[FactSales]

Keep identity: ☐ Keep nulls: ☐ Table lock: ☒ Check constraints: ☒

Rows per batch: 2147483647

Maximum insert commit size: 2147483647

View Existing

OK Cancel Help

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager

Mappings

Error Output

Available Input Columns

| Name           |
|----------------|
| Txn_ID         |
| Customer_id    |
| Purchase_DATE  |
| Revenue_Total  |
| N_Purchases    |
| Purchase_VALUE |
| Pay_Method     |

Available Destination Columns

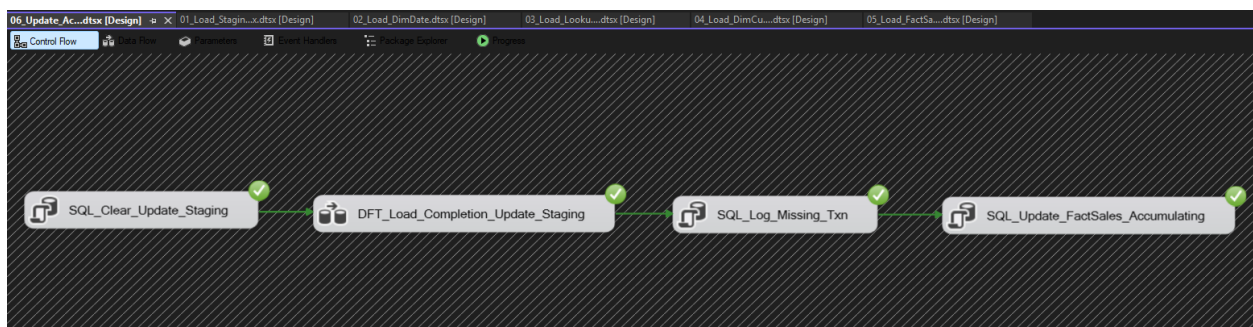
| Name         |
|--------------|
| Sales_Key    |
| Txn_ID       |
| Date_Key     |
| Customer_Key |
| Payment_Key  |
| Browser_Key  |
| Voucher_Key  |

| Input Column           | Destination Column     |
|------------------------|------------------------|
| <ignore>               | Sales_Key              |
| Txn_ID                 | Txn_ID                 |
| Date_Key               | Date_Key               |
| Customer_Key           | Customer_Key           |
| Payment_Key            | Payment_Key            |
| Browser_Key            | Browser_Key            |
| Voucher_Key            | Voucher_Key            |
| Revenue_Total          | Revenue_Total          |
| Purchase_VALUE         | Purchase_VALUE         |
| N_Purchases            | N_Purchases            |
| Time_Spent             | Time_Spent             |
| accm_bn_create_time    | accm_bn_create_time    |
| accm_bn_complete_time  | accm_bn_complete_time  |
| txn_process_time_hours | txn_process_time_hours |

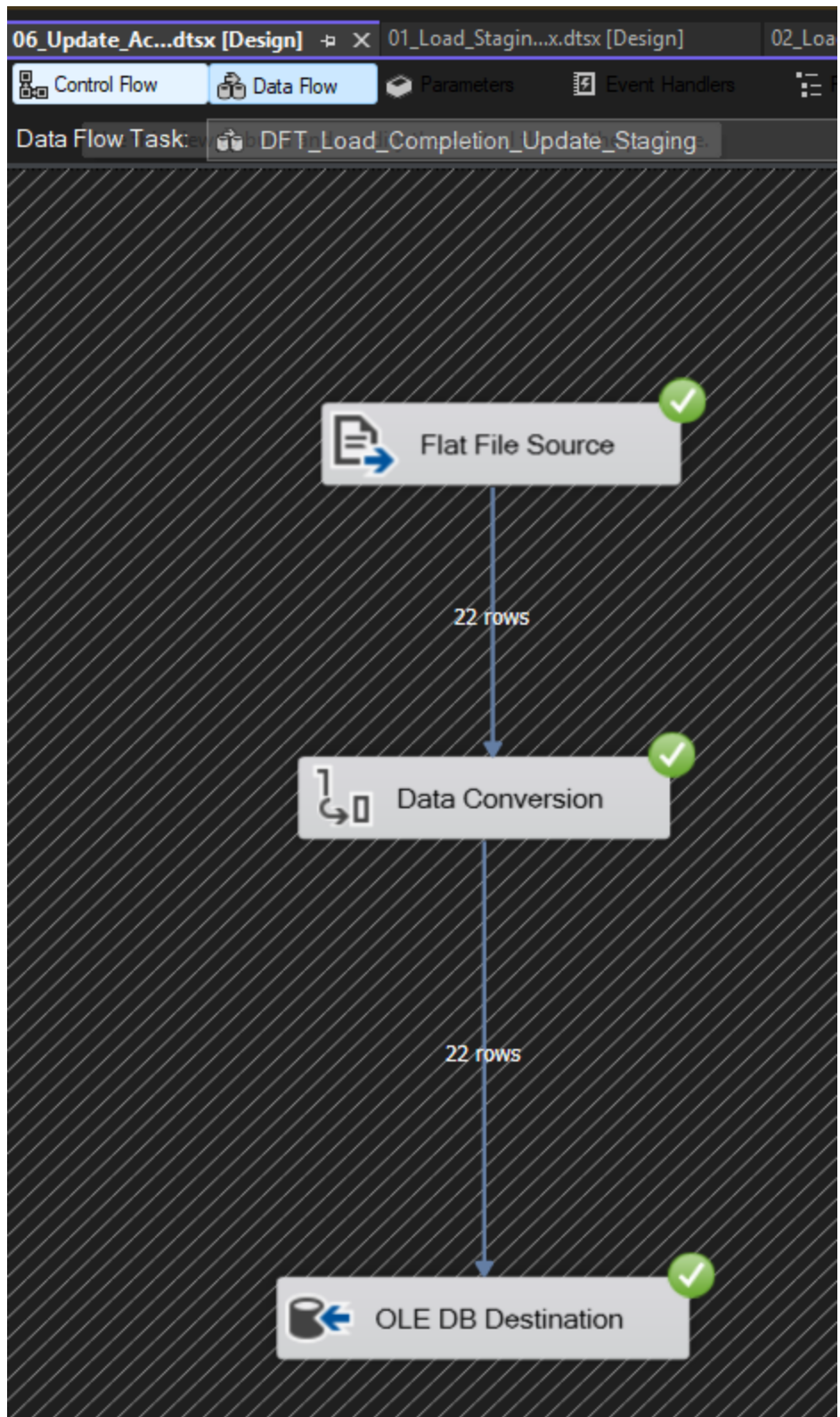
OK Cancel Help

## 6: ETL development – Accumulating fact tables

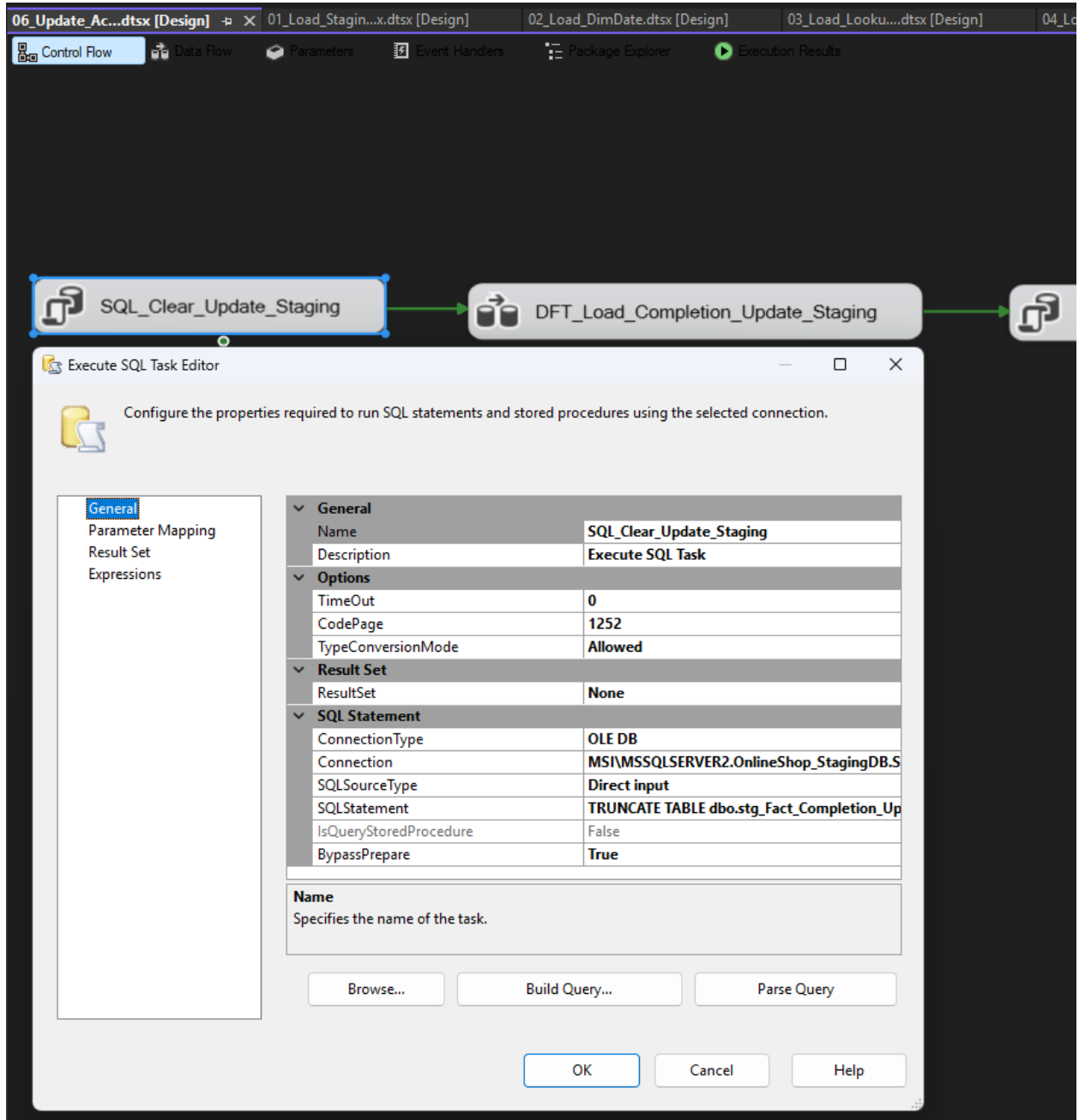
- A separate ETL SSIS package was developed to implement the accumulating fact table update process. According to the assignment requirement, the fact table was extended with the additional columns `accm_txn_create_time`, `accm_txn_complete_time`, and `txn_process_time_hours`. During the initial loading process, `accm_txn_create_time` was set to the current system date and time. Later, a separate dataset containing `Txn_ID` and `accm_txn_complete_time` was prepared and used as the source for the accumulating update.
- For this project, a separate CSV file named `fact_completion_updates.csv` was created with the columns `Txn_ID` and `accm_txn_complete_time`. This file was loaded into the staging table `stg_Fact_Completion_Update` using a separate SSIS package named `06_Update_Accumulating_Fact.dtsx`. After loading the update data into staging, rows with missing `Txn_ID` values in the fact table were identified and logged into a rejected rows table. Then the valid matching rows were used to update `FactSales`, setting `accm_txn_complete_time` and calculating `txn_process_time_hours` as the difference in hours between `accm_txn_create_time` and `accm_txn_complete_time`. This approach satisfied the requirement to implement the accumulating fact update using a separate ETL package.
- Validation queries were executed after the package ran to confirm that the fact table was updated correctly and that missing transaction IDs were handled separately. The results showed the updated completion timestamps, calculated process hours, and the rejected rows that did not have matching transaction IDs in the fact table.







- Below picture shows the configuration of the three Execute SQL Tasks and one Data Flow Task added which are named SQL\_Clear\_Update\_staging, SQL\_Log\_Missing\_Txn, SQL\_Update\_FactSales\_Accumulating and DFT\_Load\_Completion\_Update\_Staging respectively.



06\_Update\_Ac...dtss [Design]\* 01\_Load\_Stagin...x.dtsx [Design] 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCu...dtsx [Design] 05\_Load\_FactSa...

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: DFT\_Load\_Completion\_Update\_Staging

```
graph TD; A[Flat File Source] --> B[Data Conversion]; B --> C[OLE DB Destination];
```

Flat File Source Editor

Configure the properties used to connect to and obtain data from a text file.

Connection Manager Columns Error Output

Available External Columns

- ☒ Name
- ☒ Txn\_ID
- ☒ accm\_txn\_complete\_time

| External Column        | Output Column          |
|------------------------|------------------------|
| Txn_ID                 | Txn_ID                 |
| accm_txn_complete_time | accm_txn_complete_time |

OK Cancel Help

06\_Update\_Ac...dtss [Design]\* 01\_Load\_Stagin...x.dtsx [Design] 02\_Load\_DimDate.dtsx [Design] 03\_Load\_Looku...dtsx [Design] 04\_Load\_DimCu...dtsx [Design] 05\_Load\_FactSa...

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

Data Flow Task: DFT\_Load\_Completion\_Update\_Staging

```
graph TD; A[Flat File Source] --> B[Data Conversion]; B --> C[OLE DB Destination];
```

Data Conversion Transformation Editor

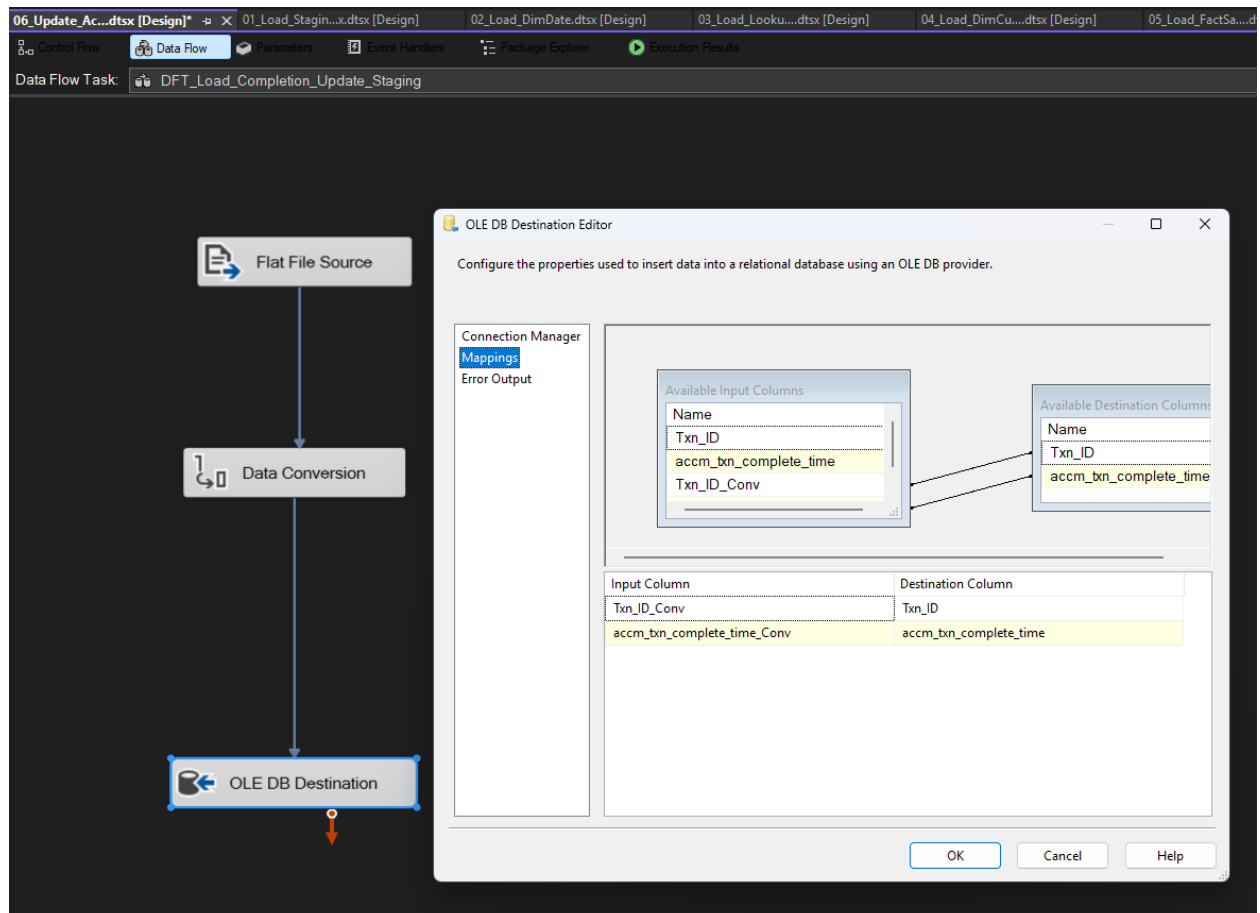
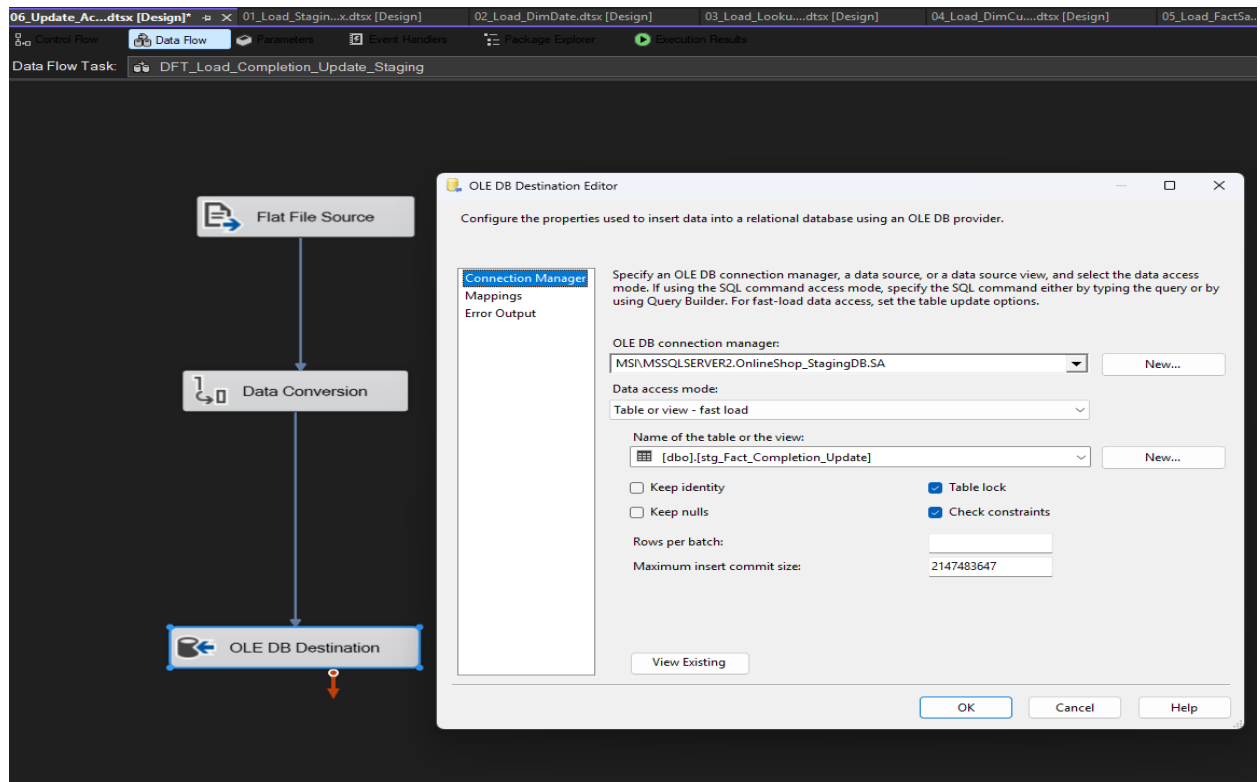
Configure the properties used to convert the data type of an input column to a different data type. Depending on the data type to which the column is converted, set the length, precision, scale, and code page of the column.

Available Input Columns

- ☒ Name
- ☒ Txn\_ID
- ☒ accm\_txn\_complete\_time

| Input Column            | Output Alias            | Data Type                    | Length | Precision | Scale | Code Page |
|-------------------------|-------------------------|------------------------------|--------|-----------|-------|-----------|
| Txn_ID                  | Txn_ID_Conv             | four-byte signed integer ... |        |           |       |           |
| accm_txn_complete_ti... | accm_txn_complete_ti... | database timestamp [DT...    |        |           |       |           |

Configure Error Output... OK Cancel Help



06\_Update\_Ac...dtss [Design]\* 01\_Load\_Stagin...x.dttss [Design] 02\_Load\_DimDate.dttss [Design] 03\_Load\_Looku...dttss [Design] 04\_Load\_DimCu...dttss [Design] 05\_Load\_FactSa...dttss [Design]

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

SQL\_Clear\_Update\_Staging DFT\_Load\_Completion\_Update\_Staging SQL\_Log\_Missing\_Txn SQL\_Update\_Fac

Execute SQL Task Editor

Configure the properties required to run SQL statements and stored procedures using the selected connection.

General

Parameter Mapping  
Result Set  
Expressions

General

|             |                     |
|-------------|---------------------|
| Name        | SQL_Log_Missing_Txn |
| Description | Execute SQL Task    |

Options

|                    |         |
|--------------------|---------|
| TimeOut            | 0       |
| CodePage           | 1252    |
| TypeConversionMode | Allowed |

Result Set

|           |      |
|-----------|------|
| ResultSet | None |
|-----------|------|

SQL Statement

|                        |                                            |
|------------------------|--------------------------------------------|
| ConnectionType         | OLE DB                                     |
| Connection             | MSI\MSSQLSERVER2.OnlineShop_StagingDB.S    |
| SQLSourceType          | Direct input                               |
| SQLStatement           | INSERT INTO OnlineShop_StagingDB.dbo.stg_F |
| IsQueryStoredProcedure | False                                      |
| BypassPrepare          | True                                       |

Name

Specifies the name of the task.

Browse... Build Query... Parse Query

OK Cancel Help

06\_Update\_Ac...dtss [Design]\* 01\_Load\_Stagin...x.dttss [Design] 02\_Load\_DimDate.dttss [Design] 03\_Load\_Looku...dttss [Design] 04\_Load\_DimCu...dttss [Design] 05\_Load\_FactSa...dttss [Design]

Control Flow Data Flow Parameters Event Handlers Package Explorer Execution Results

SQL\_Clear\_Update\_Staging DFT\_Load\_Completion\_Update\_Staging SQL\_Log\_Missing\_Txn SQL\_Update\_FactSales\_Accumulating

Execute SQL Task Editor

Configure the properties required to run SQL statements and stored procedures using the selected connection.

General

Parameter Mapping  
Result Set  
Expressions

General

|             |                                   |
|-------------|-----------------------------------|
| Name        | SQL_Update_FactSales_Accumulating |
| Description | Execute SQL Task                  |

Options

|                    |         |
|--------------------|---------|
| TimeOut            | 0       |
| CodePage           | 1252    |
| TypeConversionMode | Allowed |

Result Set

|           |      |
|-----------|------|
| ResultSet | None |
|-----------|------|

SQL Statement

|                        |                                         |
|------------------------|-----------------------------------------|
| ConnectionType         | OLE DB                                  |
| Connection             | MSI\MSSQLSERVER2.OnlineShop_DW.SA       |
| SQLSourceType          | Direct input                            |
| SQLStatement           | UPDATE fSET faccm_txn_complete_time = s |
| IsQueryStoredProcedure | False                                   |
| BypassPrepare          | True                                    |

Name

Specifies the name of the task.

Browse... Build Query... Parse Query

OK Cancel Help

- The **fact\_completion\_updates.csv** file contains transaction-level completion records, capturing each Txn\_ID along with its corresponding accm\_txn\_complete\_time timestamp. It serves as a source dataset for loading and updating fact tables in the data warehouse, ensuring accurate tracking of transaction completion events.

| Txn_ID | accm_txn_complete_time |
|--------|------------------------|
| 1      | 2026-03-31 01:08:52    |
| 2      | 2026-03-29 23:08:52    |
| 3      | 2026-03-30 14:08:52    |
| 4      | 2026-03-31 01:08:52    |
| 5      | 2026-03-30 21:08:52    |
| 6      | 2026-03-30 06:08:52    |
| 7      | 2026-03-30 09:08:52    |
| 8      | 2026-03-28 15:08:52    |
| 9      | 2026-03-31 01:08:52    |
| 10     | 2026-03-30 02:08:52    |
| 11     | 2026-03-29 01:08:52    |
| 12     | 2026-03-30 21:08:52    |
| 13     | 2026-03-29 10:08:52    |
| 14     | 2026-03-29 17:08:52    |
| 15     | 2026-03-30 22:08:52    |
| 16     | 2026-03-31 08:08:52    |
| 17     | 2026-03-30 20:08:52    |
| 18     | 2026-03-29 05:08:52    |
| 19     | 2026-03-30 03:08:52    |
| 20     | 2026-03-31 02:08:52    |
| 999999 | 2026-04-03 10:00:00    |
| 999998 | 2026-04-03 11:00:00    |

- The SQL query verifies that the **fact\_completion\_updates.csv** data has been successfully loaded into the staging table `stg_Fact_Completion_Update` by counting the total rows and previewing sample records. The results confirm that all records are correctly inserted and available for further ETL processing.

The screenshot displays the SQL Server Enterprise Manager interface. At the top, the query editor shows the following SQL code:

```
1 USE OnlineShop_StagingDB;
2 GO
3
4 SELECT COUNT(*) AS UpdateRows
5 FROM dbo_stg_Fact_Completion_Update;
6
7 SELECT TOP 20 *
8 FROM dbo_stg_Fact_Completion_Update;
9 GO
```

Below the query editor, the 'Results' tab is active, showing the output of the query. The first result is a single row with the value 22 under the column 'UpdateRows'. The second result is a table with two columns: 'Txn\_ID' and 'acom\_txn\_complete\_time', displaying the first 20 records of the staging table.

|   | UpdateRows |
|---|------------|
| 1 | 22         |

|    | Txn_ID | acom_txn_complete_time  |
|----|--------|-------------------------|
| 1  | 1      | 2026-03-31 01:08:52.000 |
| 2  | 2      | 2026-03-29 23:08:52.000 |
| 3  | 3      | 2026-03-30 14:08:52.000 |
| 4  | 4      | 2026-03-31 01:08:52.000 |
| 5  | 5      | 2026-03-30 21:08:52.000 |
| 6  | 6      | 2026-03-30 06:08:52.000 |
| 7  | 7      | 2026-03-30 09:08:52.000 |
| 8  | 8      | 2026-03-28 15:08:52.000 |
| 9  | 9      | 2026-03-31 01:08:52.000 |
| 10 | 10     | 2026-03-30 02:08:52.000 |
| 11 | 11     | 2026-03-29 01:08:52.000 |
| 12 | 12     | 2026-03-30 21:08:52.000 |
| 13 | 13     | 2026-03-29 10:08:52.000 |
| 14 | 14     | 2026-03-29 17:08:52.000 |
| 15 | 15     | 2026-03-30 22:08:52.000 |
| 16 | 16     | 2026-03-31 08:08:52.000 |
| 17 | 17     | 2026-03-30 20:08:52.000 |
| 18 | 18     | 2026-03-29 05:08:52.000 |
| 19 | 19     | 2026-03-30 03:08:52.000 |
| 20 | 20     | 2026-03-31 02:08:52.000 |

At the bottom of the interface, a yellow status bar indicates: **Query executed successfully.**

- The SQL query checks the rejected records in `stg_Fact_Completion_Update_Rejected`, confirming that invalid `Txn_IDS` intentionally added to the dataset are correctly identified and separated. The results show these records along with rejection reasons, validating that the data quality and error-handling logic in the ETL process is functioning properly.

The screenshot displays the SQL Server Enterprise Manager interface. At the top, there are two tabs: 'SQLQuery3.sq...\_DW (SA (52))' and 'SQLQuery2.s...DB (SA (77))\*'. The active tab shows a SQL query with line numbers 1 through 9. The query is as follows:

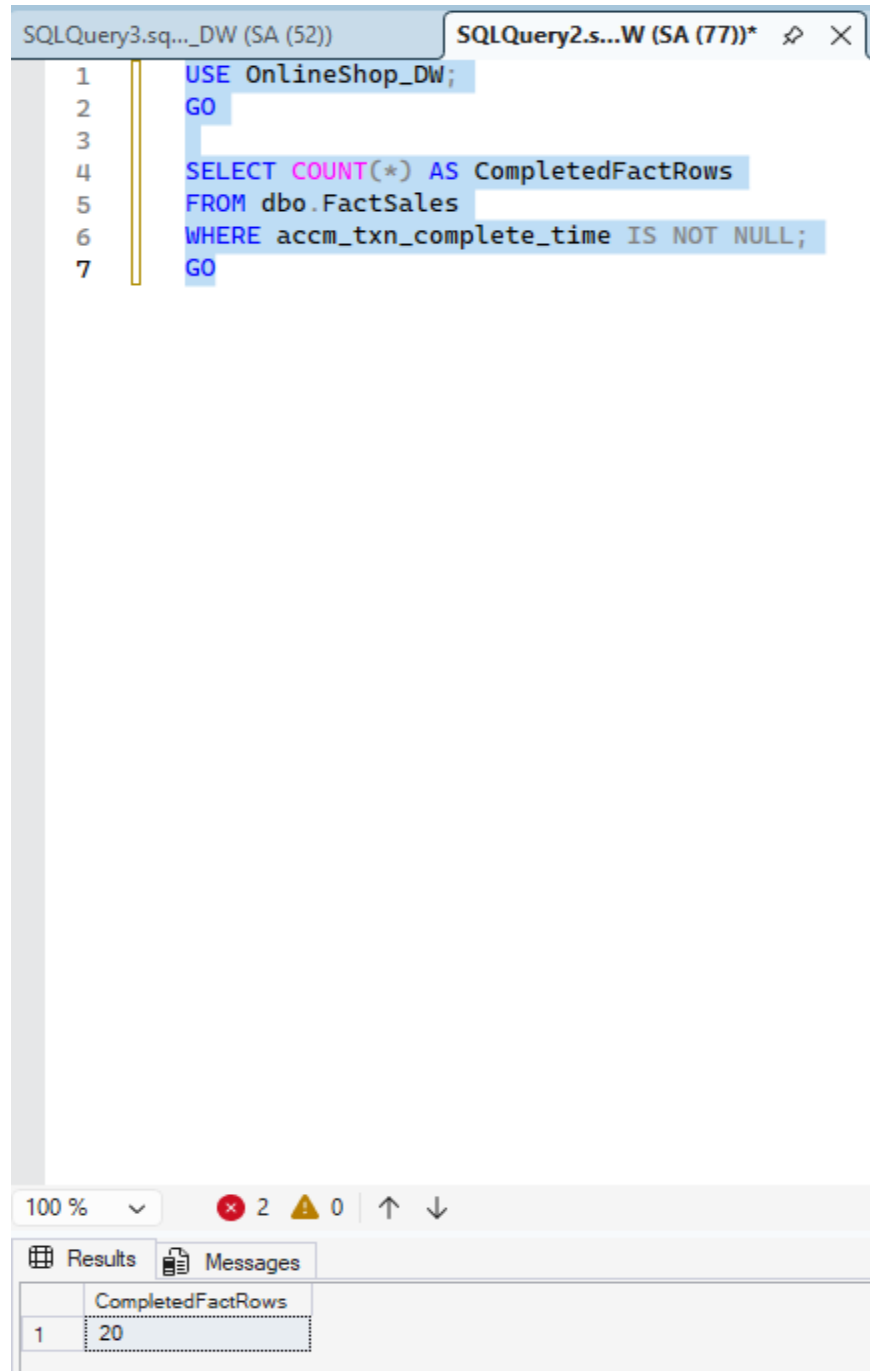
```
1  USE OnlineShop_StagingDB;  
2  GO  
3  
4  SELECT COUNT(*) AS RejectedRows  
5  FROM dbo.stg_Fact_Completion_Update_Rejected;  
6  
7  SELECT *  
8  FROM dbo.stg_Fact_Completion_Update_Rejected;  
9  GO
```

Below the query editor, the 'Results' pane is visible, showing the output of the query. It contains two tables. The first table has one column, 'RejectedRows', and one row with the value '2'. The second table has four columns: 'Txn\_ID', 'accm\_txn\_complete\_time', and 'Reject\_Reason'. It contains two rows of data:

|   | Txn_ID | accm_txn_complete_time  | Reject_Reason                 |
|---|--------|-------------------------|-------------------------------|
| 1 | 999998 | 2026-04-03 11:00:00.000 | Txn_ID not found in FactSales |
| 2 | 999999 | 2026-04-03 10:00:00.000 | Txn_ID not found in FactSales |



- The SQL query verifies that the accm\_txn\_complete\_time has been successfully updated in the FactSales table by counting records where the completion time is not null. The result confirms that valid transactions have been correctly updated in the data warehouse.



The screenshot displays the SQL Server Enterprise Manager interface. At the top, two query windows are open: 'SQLQuery3.sq...\_DW (SA (52))' and 'SQLQuery2.s...W (SA (77))\*'. The active window shows a SQL query with line numbers 1 through 7. The query is as follows:

```
1  USE OnlineShop_DW;  
2  GO  
3  
4  SELECT COUNT(*) AS CompletedFactRows  
5  FROM dbo.FactSales  
6  WHERE accm_txn_complete_time IS NOT NULL;  
7  GO
```

Below the query window, the 'Results' tab is selected, showing a single row of data. The column is named 'CompletedFactRows' and the value is 20. The status bar at the bottom indicates 100% zoom, 2 errors, and 0 warnings.

|   | CompletedFactRows |
|---|-------------------|
| 1 | 20                |

- The SQL query retrieves sample records from the FactSales table to verify that txn\_process\_time\_hours has been correctly calculated based on transaction create and completion times. The results confirm that the processing duration is accurately derived and stored for each complete transaction.

SQLQuery3.sq...\_DW (SA (52))    SQLQuery2.s...W (SA (77))\*    SQLQuery1.sq...DW

```

1  USE OnlineShop_DW;
2  GO
3
4  SELECT TOP 20
5      Txn_ID,
6      accm_txn_create_time,
7      accm_txn_complete_time,
8      txn_process_time_hours
9  FROM dbo.FactSales
10 WHERE accm_txn_complete_time IS NOT NULL
11 ORDER BY Txn_ID;
12 GO

```

100 %    2    0    ↑    ↓

Results    Messages

|    | Txn_ID | accm_txn_create_time    | accm_txn_complete_time  | txn_process_time_hours |
|----|--------|-------------------------|-------------------------|------------------------|
| 1  | 1      | 2026-03-27 13:08:52.157 | 2026-03-31 01:08:52.000 | 84                     |
| 2  | 2      | 2026-03-27 13:08:52.157 | 2026-03-29 23:08:52.000 | 58                     |
| 3  | 3      | 2026-03-27 13:08:52.157 | 2026-03-30 14:08:52.000 | 73                     |
| 4  | 4      | 2026-03-27 13:08:52.157 | 2026-03-31 01:08:52.000 | 84                     |
| 5  | 5      | 2026-03-27 13:08:52.157 | 2026-03-30 21:08:52.000 | 80                     |
| 6  | 6      | 2026-03-27 13:08:52.157 | 2026-03-30 06:08:52.000 | 65                     |
| 7  | 7      | 2026-03-27 13:08:52.157 | 2026-03-30 09:08:52.000 | 68                     |
| 8  | 8      | 2026-03-27 13:08:52.157 | 2026-03-28 15:08:52.000 | 26                     |
| 9  | 9      | 2026-03-27 13:08:52.157 | 2026-03-31 01:08:52.000 | 84                     |
| 10 | 10     | 2026-03-27 13:08:52.157 | 2026-03-30 02:08:52.000 | 61                     |
| 11 | 11     | 2026-03-27 13:08:52.157 | 2026-03-29 01:08:52.000 | 36                     |
| 12 | 12     | 2026-03-27 13:08:52.157 | 2026-03-30 21:08:52.000 | 80                     |
| 13 | 13     | 2026-03-27 13:08:52.157 | 2026-03-29 10:08:52.000 | 45                     |
| 14 | 14     | 2026-03-27 13:08:52.157 | 2026-03-29 17:08:52.000 | 52                     |
| 15 | 15     | 2026-03-27 13:08:52.157 | 2026-03-30 22:08:52.000 | 81                     |
| 16 | 16     | 2026-03-27 13:08:52.157 | 2026-03-31 08:08:52.000 | 91                     |
| 17 | 17     | 2026-03-27 13:08:52.157 | 2026-03-30 20:08:52.000 | 79                     |
| 18 | 18     | 2026-03-27 13:08:52.157 | 2026-03-29 05:08:52.000 | 40                     |
| 19 | 19     | 2026-03-27 13:08:52.157 | 2026-03-30 03:08:52.000 | 62                     |
| 20 | 20     | 2026-03-27 13:08:52.157 | 2026-03-31 02:08:52.000 | 85                     |

38

39

40

41

42

43

44

45

USE OnlineShop\_DW;

GO

SELECT \*

FROM dbo.FactSales

WHERE accm\_txn\_complete\_time IS NOT NULL

OR txn\_process\_time\_hours IS NOT NULL;

100 %

No issues found

Ln: 39, Ch: 1 (139 chars, 7 lines) SPC CRLF Windows 1252

Results

Messages

|    | Sales_Key | Txn_ID | Date_Key | Customer_Key | Payment_Key | Browser_Key | Voucher_Key | Revenue_Total | Purchase_VALUE | N_Purchases | Time_Spent | accm_txn_create_time    | accm_txn_complete_time  | txn_process_time_hours |
|----|-----------|--------|----------|--------------|-------------|-------------|-------------|---------------|----------------|-------------|------------|-------------------------|-------------------------|------------------------|
| 1  | 1097      | 1      | 20210622 | 9            | 1010        | 1005        | 1003        | 45.30         | 24.92          | 2           | 885        | 2026-03-29 11:14:25.967 | 2026-03-31 01:08:00.000 | 38                     |
| 2  | 1098      | 2      | 20211210 | 10           | 1011        | 1005        | 1004        | 36.20         | 2.90           | 3           | 656        | 2026-03-29 11:14:25.967 | 2026-03-29 23:08:00.000 | 12                     |
| 3  | 1099      | 3      | 20210314 | 11           | 1009        | 1005        | 1003        | 10.60         | 10.60          | 1           | 761        | 2026-03-29 11:14:25.967 | 2026-03-30 14:08:00.000 | 27                     |
| 4  | 1100      | 4      | 20211025 | 12           | 1010        | 1005        | 1003        | 54.10         | 43.28          | 5           | 906        | 2026-03-29 11:14:25.967 | 2026-03-31 01:08:00.000 | 38                     |
| 5  | 1101      | 5      | 20210914 | 13           | 1010        | 1005        | 1003        | 56.90         | 56.90          | 1           | 605        | 2026-03-29 11:14:25.967 | 2026-03-30 21:08:00.000 | 34                     |
| 6  | 1102      | 6      | 20210514 | 14           | 1010        | 1006        | 1003        | 13.70         | 12.47          | 6           | 364        | 2026-03-29 11:14:25.967 | 2026-03-30 06:08:00.000 | 19                     |
| 7  | 1103      | 7      | 20210109 | 15           | 1009        | 1005        | 1003        | 30.70         | 2.46           | 6           | 654        | 2026-03-29 11:14:25.967 | 2026-03-30 09:08:00.000 | 22                     |
| 8  | 1104      | 8      | 20210328 | 16           | 1012        | 1005        | 1003        | 8.10          | 6.56           | 7           | 1011       | 2026-03-29 11:14:25.967 | 2026-03-28 15:08:00.000 | -20                    |
| 9  | 1105      | 9      | 20210804 | 17           | 1009        | 1008        | 1003        | 18.00         | 11.88          | 4           | 312        | 2026-03-29 11:14:25.967 | 2026-03-31 01:08:00.000 | 38                     |
| 10 | 1106      | 10     | 20211006 | 18           | 1012        | 1005        | 1003        | 19.20         | 11.90          | 4           | 828        | 2026-03-29 11:14:25.967 | 2026-03-30 02:08:00.000 | 15                     |
| 11 | 1107      | 11     | 20211231 | 19           | 1011        | 1005        | 1004        | 36.50         | 31.39          | 5           | 1029       | 2026-03-29 11:14:25.967 | 2026-03-29 01:08:00.000 | -10                    |
| 12 | 1108      | 12     | 20211122 | 20           | 1012        | 1006        | 1003        | 14.00         | 4.34           | 4           | 479        | 2026-03-29 11:14:25.967 | 2026-03-30 21:08:00.000 | 34                     |
| 13 | 1109      | 13     | 20210802 | 21           | 1012        | 1005        | 1003        | 14.70         | 2.94           | 2           | 645        | 2026-03-29 11:14:25.967 | 2026-03-29 10:08:00.000 | -1                     |
| 14 | 1110      | 14     | 20210507 | 22           | 1012        | 1006        | 1003        | 37.40         | 15.33          | 4           | 501        | 2026-03-29 11:14:25.967 | 2026-03-29 17:08:00.000 | 6                      |
| 15 | 1111      | 15     | 20210502 | 23           | 1012        | 1007        | 1003        | 15.40         | 15.40          | 1           | 802        | 2026-03-29 11:14:25.967 | 2026-03-30 22:08:00.000 | 35                     |
| 16 | 1112      | 16     | 20210604 | 24           | 1012        | 1005        | 1003        | 28.70         | 0.86           | 7           | 804        | 2026-03-29 11:14:25.967 | 2026-03-31 08:08:00.000 | 45                     |
| 17 | 1113      | 17     | 20210222 | 25           | 1011        | 1008        | 1003        | 39.70         | 30.57          | 3           | 931        | 2026-03-29 11:14:25.967 | 2026-03-30 20:08:00.000 | 33                     |
| 18 | 1114      | 18     | 20210713 | 26           | 1012        | 1008        | 1003        | 5.10          | 1.53           | 3           | 911        | 2026-03-29 11:14:25.967 | 2026-03-29 05:08:00.000 | -6                     |
| 19 | 1115      | 19     | 20210913 | 27           | 1010        | 1005        | 1004        | 43.90         | 19.75          | 6           | 468        | 2026-03-29 11:14:25.967 | 2026-03-30 03:08:00.000 | 16                     |
| 20 | 1116      | 20     | 20210116 | 28           | 1011        | 1005        | 1003        | 36.40         | 14.56          | 6           | 714        | 2026-03-29 11:14:25.967 | 2026-03-31 02:08:00.000 | 39                     |

Query executed successfully.

MSI\MSSQLSERVER2 (17.0 RTM) SA (101) OnlineShop\_DW 00:00:00 Row: 1, Col: 1 20 rows

